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Turbine engine with composite airfoils

Abstract

A turbine engine with an engine core defining an engine centerline and comprising a rotor and a stator. The turbine engine including a set of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor. An airfoil in the set of composite airfoils including a composite portion extending chordwise between a composite leading edge and a trailing edge and a leading edge protector coupled to the composite portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation-in-part of U.S. application Ser. No. 18/741,419, filed Jun. 12, 2024, which is a continuation-in-part of U.S. application Ser. No. 18/171,533, filed Feb. 20, 2023, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD
(1) The disclosure generally relates to a component for a turbine engine, more specifically, to a composite airfoil.
BACKGROUND
(2) Composite materials typically include a fiber-reinforced matrix and exhibit a high strength-to-weight ratio. Due to the strength-to-weight ratio and moldability to adopt relatively complex shapes, composite materials are utilized in various applications, such as a turbine engine or an aircraft. Composite materials can be, for example, installed on or define a portion of the fuselage and/or wings, rudder, manifold, airfoil, or other components of the aircraft or turbine engine. Extreme loading or sudden forces can be applied to the composite components of the aircraft or turbine engine. For example, extreme loading can occur to one or more airfoils during ingestion of various materials by the turbine engine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:
(2) FIG. **1** is a schematic cross-sectional view of a turbine engine in accordance with an exemplary embodiment of the present disclosure.
(3) FIG. **2** is a schematic illustration of a composite airfoil in the form of a fan blade for the turbine engine of FIG. **1** according to an exemplary embodiment of the present disclosure.
(4) FIG. **3** is a schematic cross-section taken along line III-III of FIG. **2**.
(5) FIG. **4** is a schematic enlarged view of an exemplary fan section for the turbine engine of FIG. **1** according to an exemplary embodiment of the present disclosure.
(6) FIG. **5** is a schematic cross-sectional view of a gas turbine engine in accordance with another exemplary embodiment of the present disclosure.
(7) FIG. **6** is a schematic cross-sectional view of a gas turbine engine in accordance with another exemplary embodiment of the present disclosure.
(8) FIG. **7** is an enlarged cross-sectional view of one of the fan platforms of the present disclosure.
(9) FIG. **8** is a bottom view of the fan platform of FIG. **7**.
(10) FIG. **9** is an axial cross-sectional view of the fan platform of FIG. **7**.
(11) FIG. **10** is an exploded perspective view illustration of a fan rotor having a non-integral fan platform.
(12) FIG. **11** is a cross-sectional view illustration of the fan disk rotor illustrated in FIG. **10** with the non-integral fan platform secured thereto.
(13) FIG. **12** is a perspective view of the non-integral fan platform illustrated in FIG. **10**.
DETAILED DESCRIPTION

(14) Aspects of the disclosure herein are directed to a plurality of composite airfoil stages. For purposes of illustration, the present disclosure will be described with respect to the plurality of composite airfoil stages within an engine being a first stage of airfoils in the form of fan blades and a second stage of airfoils immediately downstream the first stage of airfoils as an outlet guide vane (OGV). While fan blades and OGVs are illustrated, it should be understood that any consecutive sets of stages are contemplated. Further, it will be understood, that aspects of the disclosure herein are not so limited and may have general applicability within an engine, including compressors, as well as in non-aircraft applications, such as other mobile applications and non-mobile industrial, commercial, and residential applications.
(15) Reference will now be made in detail to composite fan blades and composite OGVs, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings.
(16) The term "composite," as used herein is, is indicative of a material that does not include metal material. A composite can be a combination of at least two or more non-metallic elements or materials. Examples of a composite material can be, but not limited to, a polymer matrix composite (PMC), a ceramic matrix composite (CMC), carbon fibers, a polymeric resin, a thermoplastic, bismaleimide (BMI), a polyimide materials, an epoxy resin, glass fibers, and silicon matrix materials.
(17) As used herein, a "composite" component refers to a structure or a component including any suitable composite material. Composite components, such as a composite airfoil, can include several layers or plies of composite material. The layers or plies can vary in stiffness, material, and dimension to achieve the desired composite component or composite portion of a component having a predetermined weight, size, stiffness, and strength.
(18) One or more layers of adhesive can be used in forming or coupling composite components. Adhesives can include resin and phenolics, wherein the adhesive can require curing at elevated temperatures or other hardening techniques.
(19) As used herein, PMC refers to a class of materials. By way of example, the PMC material is defined in part by a prepreg, which is a reinforcement material pre-impregnated with a polymer matrix material, such as thermoplastic resin. Non-limiting examples of processes for producing thermoplastic prepregs include hot melt pre-pregging in which the fiber reinforcement material is drawn through a molten bath of resin and powder pre-pregging in which a resin is deposited onto the fiber reinforcement material, by way of non-limiting example electrostatically, and then adhered to the fiber, by way of non-limiting example, in an oven or with the assistance of heated rollers. The prepregs can be in the form of unidirectional tapes or woven fabrics, which are then stacked on top of one another to create the number of stacked plies desired for the part.
(20) Multiple layers of prepreg are stacked to the proper thickness and orientation for the composite component and then the resin is cured and solidified to render a fiber reinforced composite part. Resins for matrix materials of PMCs can be generally classified as thermosets or thermoplastics. Thermoplastic resins are generally categorized as polymers that can be repeatedly softened and flowed when heated and hardened when sufficiently cooled due to physical rather than chemical changes. Notable example classes of thermoplastic resins include nylons, thermoplastic polyesters, polyaryletherketones, and polycarbonate resins. Specific example of high performance thermoplastic resins that have been contemplated for use in aerospace applications include, polyetheretherketone (PEEK), polyetherketoneketone (PEKK), polyetherimide (PEI), polyaryletherketone (PAEK), and polyphenylene sulfide (PPS). In contrast, once fully cured into a hard rigid solid, thermoset resins do not undergo significant softening when heated, but instead thermally decompose when sufficiently heated. Notable examples of thermoset resins include epoxy, bismaleimide (BMI), and polyimide resins.
(21) Instead of using a prepreg, in another non-limiting example, with the use of thermoplastic polymers, it is possible to utilize a woven fabric. Woven fabric can include, but is not limited to,

dry carbon fibers woven together with thermoplastic polymer fibers or filaments. Non-prepreg braided architectures can be made in a similar fashion. With this approach, it is possible to tailor the fiber volume of the part by dictating the relative concentrations of the thermoplastic fibers and reinforcement fibers that have been woven or braided together. Additionally, different types of reinforcement fibers can be braided or woven together in various concentrations to tailor the properties of the part. For example, glass fibers, carbon fibers, and thermoplastic fibers could all be woven together in various concentrations to tailor the properties of the part. The carbon fibers provide the strength of the system, the glass fibers can be incorporated to enhance the impact properties, which is a design characteristic for parts located near the inlet of the engine, and the thermoplastic fibers provide the binding for the reinforcement fibers.
(22) In yet another non-limiting example, resin transfer molding (RTM) can be used to form at least a portion of a composite component. Generally, RTM includes the application of dry fibers or matrix material to a mold or cavity. The dry fibers or matrix material can include prepreg, braided material, woven material, or any combination thereof.
(23) Resin can be pumped into or otherwise provided to the mold or cavity to impregnate the dry fibers or matrix material. The combination of the impregnated fibers or matrix material and the resin are then cured and removed from the mold. When removed from the mold, the composite component can require post-curing processing.
(24) It is contemplated that RTM can be a vacuum assisted process. That is, the air from the cavity or mold can be removed and replaced by the resin prior to heating or curing. It is further contemplated that the placement of the dry fibers or matrix material can be manual or automated.
(25) The dry fibers or matrix material can be contoured to shape the composite component or direct the resin. Optionally, additional layers or reinforcing layers of material differing from the dry fiber or matrix material can also be included or added prior to heating or curing.
(26) As used herein, CMC refers to a class of materials with reinforcing fibers in a ceramic matrix. Generally, the reinforcing fibers provide structural integrity to the ceramic matrix. Some examples of reinforcing fibers can include, but are not limited to, non-oxide silicon-based materials (e.g., silicon carbide, silicon nitride, or mixtures thereof), non-oxide carbon-based materials (e.g., carbon), oxide ceramics (e.g., silicon oxycarbides, silicon oxynitrides, aluminum oxide (Al.sub.2O.sub.3), silicon dioxide (SiO.sub.2), aluminosilicates such as mullite, or mixtures thereof), or mixtures thereof.
(27) Some examples of ceramic matrix materials can include, but are not limited to, non-oxide silicon-based materials (e.g., silicon carbide, silicon nitride, or mixtures thereof), oxide ceramics (e.g., silicon oxycarbides, silicon oxynitrides, aluminum oxide (Al.sub.2O.sub.3), silicon dioxide (SiO.sub.2), aluminosilicates, or mixtures thereof), or mixtures thereof. Optionally, ceramic particles (e.g., oxides of Si, Al, Zr, Y, and combinations thereof) and inorganic fillers (e.g., pyrophyllite, wollastonite, mica, talc, kyanite, and montmorillonite) can also be included within the ceramic matrix.
(28) Generally, particular CMCs can be referred to as their combination of type of fiber/type of matrix. For example, C/SiC for carbon-fiber-reinforced silicon carbide, SiC/SiC for silicon carbide-fiber-reinforced silicon carbide, SiC/SiN for silicon carbide fiber-reinforced silicon nitride, SiC/SiC—SiN for silicon carbide fiber-reinforced silicon carbide/silicon nitride matrix mixture, etc. In other examples, the CMCs can be comprised of a matrix and reinforcing fibers comprising oxide-based materials such as aluminum oxide (Al.sub.2O.sub.3), silicon dioxide (SiO.sub.2), aluminosilicates, and mixtures thereof. Aluminosilicates can include crystalline materials such as mullite (3Al2O.sub.32SiO.sub.2), as well as glassy aluminosilicates.
(29) In certain non-limiting examples, the reinforcing fibers may be bundled and/or coated prior to inclusion within the ceramic matrix. For example, bundles of the fibers may be formed as a reinforced tape, such as a unidirectional reinforced tape. A plurality of the tapes may be laid up together to form a preform component. The bundles of fibers may be impregnated with a slurry

composition prior to forming the preform or after formation of the preform. The preform may then undergo thermal processing, and subsequent chemical processing, to arrive at a component formed of a CMC material having a desired chemical composition. For example, the preform may undergo a cure or burn-out to yield a high char residue in the preform, and subsequent melt-infiltration with silicon, or a cure or pyrolysis to yield a silicon carbide matrix in the preform, and subsequent chemical vapor infiltration with silicon carbide. Additional steps may be taken to improve densification of the preform, either before or after chemical vapor infiltration, by injecting it with a liquid resin or polymer followed by a thermal processing step to fill the voids with silicon carbide. CMC material as used herein may be formed using any known or hereinafter developed methods including but not limited to melt infiltration, chemical vapor infiltration, polymer impregnation pyrolysis (PIP), or any combination thereof.

(30) Such materials, along with certain monolithic ceramics (i.e., ceramic materials without a reinforcing material), are particularly suitable for higher temperature applications. Additionally, these ceramic materials are lightweight compared to superalloys, yet can still provide strength and durability to the component made therefrom. Therefore, such materials are currently being considered for many gas turbine components used in higher temperature sections of gas turbine engines, such as airfoils (e.g., turbines, and vanes), combustors, shrouds and other like components, that would benefit from the lighter-weight and higher temperature capability these materials can offer.

(31) The term “metallic” as used herein is indicative of a material that includes metal such as, but not limited to, titanium, iron, aluminum, stainless steel, and nickel alloys. A metallic material or alloy can be a combination of at least two or more elements or materials, where at least one is a metal.

(32) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

(33) As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

(34) As used herein, the term “upstream” refers to a direction that is opposite the fluid flow direction, and the term “downstream” refers to a direction that is in the same direction as the fluid flow. The term “fore” or “forward” means in front of something and “aft” or “rearward” means behind something. For example, with regard to a turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.

(35) The term “fluid” may be a gas or a liquid, or multi-phase. The term “fluid communication” means that a fluid is capable of making the connection between the areas specified.

(36) Additionally, as used herein, the terms “radial” or “radially” refer to a direction away from a common center. For example, in the overall context of a turbine engine, radial refers to a direction along a ray extending between a center longitudinal axis of the engine and an outer engine circumference.

(37) All directional references (e.g., radial, axial, proximal, distal, upper, lower, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, clockwise, counterclockwise, upstream, downstream, forward, aft, etc.) are only used for identification purposes to aid the reader's understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of aspects of the disclosure described herein. Connection references (e.g., attached, coupled, connected, and joined) are to be construed broadly and can include intermediate structural elements between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do

not necessarily infer that two elements are directly connected and in fixed relation to one another. The exemplary drawings are for purposes of illustration only and the dimensions, positions, order and relative sizes reflected in the drawings attached hereto can vary.
(38) The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. Furthermore, as used herein, the term “set” or a “set” of elements can be any number of elements, including only one.
(39) Leading length or “LL” as used herein refers to a length between a leading edge of the airfoil and a seam between a leading edge protector and a portion of the airfoil.
(40) First leading length or “FLL” as used herein refers to the leading length of a first stage of airfoils.
(41) Second leading length or “SLL” as used herein refers to the leading length of a second stage of airfoils immediately downstream from the first stage of airfoils.
(42) Chord length or “CL” as used herein refers to a length between a leading edge of the airfoil and a trailing edge of the airfoil.
(43) First chord length or “FCL” as used herein refers to the chord length of the first stage of airfoils.
(44) Second chord length or “SCL” as used herein refers to the chord length of the second stage of airfoils.
(45) Airfoil protection factor or “APF” as used herein refers to a relationship in the form of a ratio of the leading length to the chord length of the airfoil. As more protection is provided for any given airfoil, the leading length increases and in turn so does the APF.
(46) Stage performance factor or “SPF” as used herein refers to a relationship in the form of a ratio of the airfoil protection factor for the first stage of airfoils, or “APF1” to the airfoil protection factor for the second stage of airfoils, or “APF2”.
(47) Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.
(48) In certain exemplary embodiments of the present disclosure, a turbine engine defining a centerline and a circumferential direction is provided. The turbine engine may generally include a turbomachine and a rotor assembly. The rotor assembly may be driven by the turbomachine. The turbomachine, the rotor assembly, or both may define a substantially annular flow path relative to the centerline of the turbine engine. In certain aspects of the present disclosure, an unducted or open rotor turbine engine includes a set of circumferentially spaced fan blades, which extend, exteriorly, beyond a nacelle encasing or engine core.
(49) The turbine engine includes airfoils in the form of blades and vanes. The airfoils described herein can be a plurality of airfoils provided circumferentially about the centerline or be partially provided about a portion of the centerline. At least one airfoil in the plurality of airfoils includes a protective covering on a leading edge of the airfoil. The protective covering can be a metal covering. The protective covering is referred to herein as leading edge protector.
(50) The leading edge protector can be designed for various flight conditions, including take off, descent, and idle. The objective, when designing an airfoil, specifically a composite fan blade and a composite outlet guide vane can be generally stated as balancing an added weight component from the protective covering, or sheath, on the leading edge with an acceptable amount of protection of the leading edge. The balancing of efficient weight designs can be particularly important in large turbofan applications of traditional direct drive, gear-reduction designs, and open-rotor designs. Key factors to consider include that the ratio of the leading edge chord to the blade chord is a balance between the leading edge dominating the response to a bird ingestion or similar event, and the PMC airfoil dominating the characteristics of the blade aerodynamics in normal operation.
(51) There is a tradeoff between the percent of the airfoil chord that is covered by the leading edge

protector, and the performance of the airfoil. The protective covering provides a stiffness to the airfoil for bird ingestion, but the remainder of the blade is desirable to be flexible for aerodynamic purposes. Because the fan blade rotates and the OGV is stationary the dynamics of a bird ingestion event differs for the two airfoils.

(52) The inventors have determined that the leading edge protector must overlap with enough of the composite airfoil in order to provide a strong enough bond, but it is desirable to minimize the overlap in order for the composite blade to flex. The leading edge protector also provides erosion protection to a composite airfoil and is required for both static and rotating airfoils. The leading edge protector characteristics have been developed from multiple tests and simulation analyses covering the ingestion of birds of varying sizes at varying span positions, and analysis of blades that have been returned for repair following bird strikes in revenue service. Furthermore, the OGV is not rotating, and experiences a different stress when impacted by a bird after it has passed through the fan. Multiple simulations and analyses depending on how the bird strikes the fan, whether it hits directly centered on a leading edge protector or hits between two adjacent blades where it is more likely to pass through without being sliced into smaller pieces. The relationship between the percent chord of a rotating and non-rotating blade that is covered by the leading edge protector is not obvious due to the difference in the forces acting upon the airfoils when struck by a bird or similar object.

(53) The inventors' practice has proceeded in the manner of designing airfoil stages, modifying the airfoil stages with the addition of the leading edge protector, and redesigning the airfoil stages with the leading edge protector meeting protection requirements associated with the airfoil stages. After calculating and checking an amount of protection provided and an amount of weight increase or decreases associated with the leading edge protector, the process is repeated for other stages of during the design of several different types of turbomachines, such as those shown in FIG. **1**. In other words, an airfoil design can meet performance requirements for one location in the engine, but not necessarily for another location.

(54) FIG. **1** is a schematic cross-sectional diagram of a turbine engine **10** for an aircraft. By way of non-limiting example the turbine engine **10** is illustrated as an open rotor turbine engine. The turbine engine **10** has a generally longitudinally extending axis or engine centerline **12** extending from a forward end **14** to an aft end **16**. The turbine engine **10** includes, in downstream serial flow relationship, a set of circumferentially spaced blades or propellers defining a fan section **18** including a fan **20**, a compressor section **22** including a booster or low pressure (LP) compressor **24** and a high pressure (HP) compressor **26**, a combustion section **28** including a combustor **30**, a turbine section **32** including a HP turbine **34**, and a LP turbine **36**, and an exhaust section **38**. The turbine engine **10** as described herein is meant as a non-limiting example, and other architectures are possible, such as, but not limited to, a steam turbine engine, a supercritical carbon dioxide turbine engine, or any other suitable turbine engine.

(55) An exterior surface, defined by a nacelle **40**, of the turbine engine **10** extends from the forward end **14** of the turbine engine **10** toward the aft end **16** of the turbine engine **10** and covers at least a portion of the compressor section **22**, the combustion section **28**, the turbine section **32**, and the exhaust section **38**. The fan section **18** can be positioned at a forward portion of the nacelle **40** and extend radially outward from the nacelle **40** of the turbine engine **10**, specifically, the fan section **18** extends radially outward from the nacelle **40**. The fan section **18** includes a set of fan blades **42**, and a set of outlet guide vanes (OGV) **82** downstream the set of fan blades **42**, both disposed radially about the engine centerline **12**. The turbine engine **10** includes any number of one or more sets of rotating blades or propellers (e.g., the set of fan blades **42**) disposed upstream of a set of stationary fan vanes (e.g., the set of OGVs **82**). As a non-limiting example, the turbine engine **10** can include multiple sets of rotating blades and stationary vanes. The set of fan blades **42** can include a first leading edge protector **140***a* and the set of OGVs **82** can include a second leading edge protector **140***b*. As such, the turbine engine **10** is further defined as an unducted single-fan

turbine engine. The turbine engine **10** is further defined by the location of the fan section **18** with respect to the combustion section **28**. The fan section **18** can be upstream, downstream, or in-line with the axial positioning of the combustion section **28**. In some aspects of the disclosure herein, the turbine engine can include a fan casing **80** (shown in dotted line) surrounding the fan **20** to define a ducted turbine engine.

(56) In one approach, described in greater detail below, a plurality of discrete platforms **534** are provided between the fan blades **42**.

(57) The compressor section **22**, the combustion section **28**, and the turbine section **32** are collectively referred to as an engine core **44**, which generates combustion gases. The engine core **44** is surrounded by an engine casing **46**, which is operatively coupled with a portion of the nacelle **40** of the turbine engine **10**.

(58) A HP shaft or spool **48** disposed coaxially about the engine centerline **12** of the turbine engine **10** drivingly connects the HP turbine **34** to the HP compressor **26**. A LP shaft or spool **50**, which is disposed coaxially about the engine centerline **12** of the turbine engine **10** within the larger diameter annular HP spool **48**, drivingly connects the LP turbine **36** to the LP compressor **24** and fan **20**. The HP spools **48** and the LP spool **50** are rotatable about the engine centerline **12** and couple to a set of rotatable elements, which collectively define a rotor **51**.

(59) It will be appreciated that the turbine engine **10** is either a direct drive or an integral drive engine utilizing a reduction gearbox coupling the LP shaft or spool **50** to the fan **20**.

(60) The LP compressor **24** and the HP compressor **26**, respectively, include a set of compressor stages **53**, **54**, in which a set of compressor blades **57**, **58** rotate relative to a corresponding set of static compressor vanes **60**, **62** (also called a nozzle) to compress or pressurize the stream of fluid passing through the stage. In a single compressor stage **53**, **54**, multiple compressor blades **57**, **58** are provided in a ring and extend radially outward relative to the engine centerline **12**, from a blade platform to a blade tip, while the corresponding static compressor vanes **60**, **62** are positioned upstream of and adjacent to the compressor blades **57**, **58**. It is noted that the number of blades, vanes, and compressor stages shown in FIG. **1** were selected for illustrative purposes only, and that other numbers are possible.

(61) The compressor blades **57**, **58** for a stage of the compressor are mounted to a disk **61**, which is mounted to the corresponding one of the HP and LP spools **48**, **50**, with each stage having its own disk **61**. The static compressor vanes **60**, **62** for a stage of the compressor are mounted to the engine casing **46** in a circumferential arrangement.

(62) The HP turbine **34** and the LP turbine **36**, respectively, include a set of turbine stages **64**, **66**, in which a set of turbine blades **68**, **70** are rotated relative to a corresponding set of static turbine vanes **72**, **74** (also called a nozzle) to extract energy from the stream of fluid passing through the stage. In a single turbine stage **64**, **66**, multiple turbine blades **68**, **70** are provided in a ring and extends radially outward relative to the engine centerline **12**, from a blade platform to a blade tip, while the corresponding static turbine vanes **72**, **74** are positioned upstream of and adjacent to the turbine blades **68**, **70**. It is noted that the number of blades, vanes, and turbine stages shown in FIG. **1** were selected for illustrative purposes only, and that other numbers are possible.

(63) The turbine blades **68**, **70** for a stage of the turbine are mounted to a disk **71**, which is mounted to the corresponding one of the HP and LP spools **48**, **50**, with each stage having a dedicated disk **71**. The static turbine vanes **72**, **74** for a stage of the compressor are be mounted to the engine casing **46** in a circumferential arrangement. The compressor blades **57**, **58** and the turbine blades **68**, **70** described herein can be part of a blisk, rather than being mounted to a disk.

(64) Complementary to the rotor portion, the stationary portions of the turbine engine **10**, such as the static vanes **60**, **62**, **72**, **74** among the compressor section **22** and the turbine section **32** are also referred to individually or collectively as a stator **63**. As such, the stator **63** refers to the combination of non-rotating elements throughout the turbine engine **10**.

(65) The nacelle **40** is operatively coupled to the turbine engine **10** and covers at least a portion of

the engine core **44**, the engine casing **46**, or the exhaust section **38**. At least a portion of the nacelle **40** extends axially forward or upstream the illustrated portion. For example, the nacelle **40** extends axially forward such that a portion of the nacelle **40** overlays or covers a portion of the fan section **18** or a booster section (not illustrated) of the turbine engine **10**. A pylon **84** mounts the turbine engine **10** to an exterior structure (e.g., a fuselage of an aircraft, a wing, a tail wing, etc.).
(66) During operation of the turbine engine **10**, a freestream airflow **79** flows against a forward portion of the turbine engine **10**. A portion of the freestream airflow **79** becomes an inlet airflow **78**. The inlet airflow **78** flows through the set of fan blades **42** and over the nacelle **40** of the turbine engine **10**. Subsequently, the inlet airflow **78** flows over at least a portion of the set of OGVs **82**, which directs the inlet airflow **78** such that it is transverse toward the engine centerline **12**. The inlet airflow **78** then flows past the set of OGVs **82**, following the curvature of the nacelle **40** and toward the exhaust section **38**.
(67) A portion of the freestream airflow **79** enters the engine core **44** after flowing through the set of fan blades **42** and is described as a working airflow **76**, which is used for combustion within the engine core **44**. More specifically, the working airflow **76** flows into the LP compressor **24**, which then pressurizes the working airflow **76** thus defining a pressurized airflow that is supplied to the HP compressor **26**, which further pressurizes the air. The working airflow **76**, or the pressurized airflow, from the HP compressor **26** is mixed with fuel in the combustor **30** and ignited, thereby generating combustion gases. Some work is extracted from these gases by the HP turbine **34**, which drives the HP compressor **26**. The combustion gases are discharged into the LP turbine **36**, which extracts additional work to drive the LP compressor **24**, and the working airflow **76**, or exhaust gas, is ultimately discharged from the turbine engine **10** via the exhaust section **38**. The driving of the LP turbine **36** drives the LP spool **50** to rotate the fan **20** and the LP compressor **24**. The working airflow **76**, including the pressurized airflow and the combustion gases, defines a working airflow that flows through the compressor section **22**, the combustion section **28**, and the turbine section **32** of the turbine engine **10**.
(68) The working airflow **76** and at least some of the inlet airflow **78** merge downstream of the exhaust section **38** of the turbine engine **10**. The working airflow **76** and the inlet airflow **78**, together, form an overall thrust of the turbine engine **10**.
(69) It is contemplated that a portion of the working airflow **76** is drawn as bleed **77** air (e.g., from the compressor section **22**). The bleed air **77** provides an airflow to engine components requiring cooling. The temperature of the working airflow **76** exiting the combustor **30** is significantly increased with respect to the working airflow **76** within the compressor section **22**. As such, cooling provided by the bleed air **77** is necessary for operating of such engine components in the heightened temperature environments or a hot portion of the turbine engine **10**. In the context of a turbine engine, the hot portions of the engine are normally downstream of the combustor **30**, especially the turbine section **32**, with the HP turbine **34** being the hottest portion as it is directly downstream of the combustion section **28**. Other sources of cooling fluid are, but are not limited to, fluid discharged from the LP compressor **24** or the HP compressor **26**.
(70) FIG. **2** is schematic illustration of a composite airfoil **130** in the form of, by way of non-limiting example, a fan blade **131**. The fan blade **131** can be, by way of non-limiting example, a blade of the set of fan blades **42** or a blade from the compressor blades **57**, **58** or the turbine blades **68**, **70**. Further, the composite airfoil **130** can be a vane of the set of OGVs **82** or a vane of the static vanes **60**, **62**, **72**, **74**. It is contemplated that the composite airfoil **130** can be a blade, vane, airfoil, or other component of any turbine engine, such as, but not limited to, a gas turbine engine, a turboprop engine, a turboshaft engine, or a turbofan engine.
(71) The composite airfoil **130** can include a wall **132** bounding an interior **133**. The wall **132** can define an exterior surface **134** extending radially between a leading edge **135** and a trailing edge **136** to define a chordwise direction (denoted “C”). The composite airfoil **130** has a chord length (denoted “CL”) measured along the chordwise direction C between the leading edge **135** and the

trailing edge **136**. The exterior surface **134** can further extend between a root **137** and a tip **138** to define a spanwise direction (denoted "S"). The composite airfoil **130** has a span length (denoted "SL") measured along the spanwise direction S between the root **137** and the tip **138** where the root is considered 0% of the span length SL and the tip **138** is considered 100% of the span length SL. The span length SL is the maximum distance between the root **137** and the tip **138** of the composite airfoil **130**. It will be understood that the composite airfoil **130** can take any suitable shape, profile, or form including that the leading edge **135** need not be curved.

(72) An axial direction (denoted "A") extends generally across the page from right to left. The axial direction A is parallel to the engine centerline **12** (FIG. **1**). A radial direction (denoted "R") extends perpendicularly away from the axial direction A. It should be understood that the spanwise direction S is parallel to the radial direction R. The chordwise direction C can extend generally along the axial direction A, however with more bend in the composite airfoil **130**, it should be understood that the chordwise direction C can extend both into and out of the page and across the page from left to right.

(73) The exterior surface **134** is defined by a leading edge protector **140** and a composite portion **150**. A seam **139**, separates the leading edge protector **140** from the composite portion **150** along the exterior surface **134**. The leading edge protector **140** extends along the chordwise direction C between the leading edge **135** and the seam **139** to define a leading length (denoted "LL").

(74) The leading edge protector **140** is typically a metallic leading edge protector and can be made of, but is not limited to, steel, aluminum, refractory metals such as titanium, or superalloys based on nickel, cobalt, or iron. It should be understood that the leading edge protector **140** for the fan blade **131** can be a metallic leading edge protector while a set of stationary vanes downstream from the fan blade **131**, by way of non-limiting example the set of OGVs **82** (FIG. **1**), have the second leading edge protector **140***b* (FIG. **1**) made of a polyurethane material. Further, the leading edge protectors **140**, **140***a*, **140***b* described herein can be any suitable material such as metal, thermoplastic, or polyurethane, where both are the same, or different.

(75) The composite portion **150** can include a composite leading edge **152** spaced a distance (denoted "D") from the leading edge **135**. The composite leading edge **152** can define at least a portion of, or all of the seam **139**. It is further contemplated that at least a part of the leading edge protector **140** overlaps the composite portion **150** such that at least a portion of, illustrated in dashed line, or all of the composite leading edge **152** is located upstream from the seam **139**. In other words, the leading edge protector **140** can define a sheath **144** on the composite leading edge **152**.

(76) The composite portion **150** can be made of one or more layers of material. The one or more layers of material can be applied during the same stage or different stages of the manufacturing of the composite airfoil **130**. By way of non-limiting example, composite portion **150** can include at least a polymer matrix composite (PMC) portion or a polymeric portion. The polymer matrix composite can include, but is not limited to, a matrix of thermoset (epoxies, phenolics) or thermoplastic (polycarbonate, polyvinylchloride, nylon, acrylics) and embedded glass, carbon, steel, or Kevlar fibers.

(77) The leading edge protector **140** and the composite portion **150** can be formed by a variety of methods, including additive manufacturing, casting, electroforming, or direct metal laser melting, in non-limiting examples. As used herein, an "additively manufactured" component refers to a component formed by an additive manufacturing (AM) process, wherein the component is built layer-by-layer by successive deposition of material. AM is an appropriate name to describe the technologies that build 3D objects by adding layer-upon-layer of material, whether the material is plastic, ceramic, or metal. AM technologies can utilize a computer, 3D modeling software (Computer Aided Design or CAD), machine equipment, and layering material. Once a CAD sketch is produced, the AM equipment can read in data from the CAD file and lay down or add successive layers of liquid, powder, sheet material or other material, in a layer-upon-layer fashion to fabricate

a 3D object. It should be understood that the term "additive manufacturing" encompasses many technologies including subsets like 3D Printing, Rapid Prototyping (RP), Direct Digital Manufacturing (DDM), layered manufacturing and additive fabrication. Non-limiting examples of additive manufacturing that can be utilized to form an additively-manufactured component include powder bed fusion, vat photopolymerization, binder jetting, material extrusion, directed energy deposition, material jetting, or sheet lamination. It is also contemplated that a process utilized could include printing a negative of the part, either by a refractory metal, ceramic, or printing a plastic, and then using that negative to cast the component.

(78) It will be shown herein that a relationship between the leading length LL and the chord length CL can be referred to herein as an airfoil protection factor or simply as "APF". In other words, for any given composite airfoil **130** having a predetermined chord length CL, an amount of coverage provided by the leading edge protector **140** increases, so does the leading length LL and in turn the APF.

(79) FIG. **3** is a schematic cross-section taken along line III-III of FIG. **2**. The leading edge protector **140** is the sheath **144** with a first wall **146**, a second wall **147**, and a third wall **148** interconnecting the first wall **146** and the second wall **147**. The first wall **146**, second wall **147**, and third wall **148** of the leading edge protector **140** are oriented and shaped such that they define a generally U-shaped (or C-shaped) channel **154** therebetween. As shown in FIG. **3** and as will be discussed below, the channel **154** is sized and shaped to receive the composite leading edge **152** of the composite portion **150**. Notably, the shape of the channel **154** is shown by way of example only and the channel **154** is not limited to this specific shape and is not drawn to scale.

(80) The composite airfoil **130** can extend between a first side **156** and a second side **158**. The seam **139** can be two seams **139***c*, **139***d* at corresponding ends of the channel **154**. The leading length LL is measured from the leading edge **135** to the seam **139***d* furthest from the leading edge **135**. While illustrated at two different locations, it should be understood that the seams **139***c*, **139***d* can be located at the same leading length LL. While illustrated as rectangular blunt ends at the seam **139**, the leading edge protector **140** can taper such that the leading edge protector **140** and the composite portion **150** are flush to define the exterior surface **134**.

(81) FIG. **4** is schematic enlarged view of a fan section **118** similar to fan section **18** therefore, like parts of the fan section **118** (FIG. **1**) will be identified with like numerals increased by 100, with it being understood that the description of the like parts of the fan section **18** applies to the fan section **118**, except where noted.

(82) A set of compressor stages **153** include a set of compressor blades **157** rotating relative to a corresponding set of static compressor vanes **160**. A set of fan blades **142** define a fan section **118** including a fan **120**. The turbine engine can include a fan casing **180** surrounding the fan **120**.

(83) The set of fan blades **142** defines a first stage of airfoils **200***a* within the fan section **118** (FIG. **1**). A first airfoil **230***a* in the first stage of airfoils **200***a* is similar to the previously described airfoil **130**, therefore like parts of the first airfoil **230***a* will be identified with like numerals increased by 100 and having a notation "a" with it being understood that the description of the like parts of the airfoil **130** applies to the first airfoil **230***a*, except where noted. While only a single fan blade is shown in the cross-section it will be understood that that the set of fan blades **142** are included and spaced about the fan section **118**.

(84) The first airfoil **230***a* has a first span length (denoted "SL**1**") measured along the spanwise direction S between a first root **237***a* and a first tip **238***a* where the first root **237***a* is considered 0% of the first span length SL**1** and the first tip **238***a* is considered 100% of the first span length SL**1**. The first span length SL**1** is the maximum distance between the first root **237***a* and the first tip **238***a* of the first airfoil **230***a*.

(85) A first leading edge protector **240***a* extends along the chordwise direction C between a first leading edge **235***a* and a first seam **239***a* to define a first leading length (denoted "FLL"). The first airfoil **230***a* has a first chord length (denoted "FCL") measured along the chordwise direction C

between the first leading edge **235***a* and the first trailing edge **236***a*.

(86) A relationship between the first leading length (FLL) and the first chord length (FCL) is denoted herein with a first expression of the APF:

(87) $\mathrm{APF1} = \frac{\mathrm{FLL}}{\mathrm{FCL}}$ (1)

(88) OGVs **182** define a second stage of airfoils **200***b* downstream from the first stage of airfoils **200***a*. A second airfoil **230***b* in the second stage of airfoils **200***b* is similar to the previously described airfoil **130**, therefore like parts of the second airfoil **230***b* will be identified with like numerals increased by 100 and having a notation "b" with it being understood that the description of the like parts of the airfoil **130** applies to the second airfoil **230***b*, except where noted. The second airfoil **230***b* is located downstream from the first airfoil **230***a*. While only a single outlet guide vane **182** is shown in the cross-section it will be understood that the OGVs **182** are multiple OGVs spaced about the fan section **118**.

(89) A second leading edge protector **240***b* extends along the chordwise direction C between a second leading edge **235***b* and a second seam **239***b* to define a second leading length (denoted "SLL"). The second airfoil **230***b* has a second chord length (denoted "SCL") measured along the chordwise direction C between the second leading edge **235***b* and second trailing edge **236***b*.

(90) The second airfoil **230***b* has a second span length (denoted "SL**2**") measured along the spanwise direction S between a second root **237***b* and a second tip **238***b* where the second root **237***b* is considered 0% of the second span length SL**2** and the second tip **238***b* is considered 100% of the second span length SL**2**. The second span length SL**2** is the maximum distance between the second root **237***b* and the second tip **238***b* of the second airfoil **230***b*.

(91) The first and second leading edge protectors **240***a*, **240***b* can each define first and second sheaths **244***a*, **244***b*. An exterior surface of each airfoil **230***a*, **230***b* is defined by the corresponding leading edge protectors **240***a*, **240***b* and a corresponding composite portion **250***a*, **250***b*. The composite portions **250***a*, **250***b* can each include a corresponding composite leading edge **252***a*, **252***b* which can define at least a portion of, or all of the corresponding seams **239***a*, **239***b*.

(92) A relationship between the second leading length (SLL) and the second chord length (SCL) is denoted herein with a second expression of the APF:

(93) $\mathrm{APF2} = \frac{\mathrm{SLL}}{\mathrm{SCL}}$ (2)

(94) As will be further discussed herein, the APF describes an amount of protection coverage by the leading edge protector of any of the airfoils **130**, **230***a*, **230***b* described herein. A balance trade-off between the amount of protection and the weight gain/loss associated with any of the protector portions described herein can be expressed by an APF value of from 0.1 to 0.3, inclusive of endpoints. In other words, to satisfy protection requirements the leading edge protector described herein should protect at least 10% and up to and including 30% of the composite airfoil before becoming too heavy.

(95) The first stage of airfoils **200***a* has a first number of airfoils and the second stage of composite airfoils **200***b* has a second number of airfoils different than the first number. In other words, the consecutive stages of airfoils can vary in size and number of airfoils. Further, the first stage of composite airfoils **200***a* and the second stage of composite airfoils **200***b* can both be configured to rotate.

(96) It will be appreciated that the number, size, and configuration of the composite airfoils described herein are provided by way of example only and that in other exemplary embodiments, the composite airfoils may have any other suitable configuration including that the plurality of airfoils may be in multiple rotor stages, etc.

(97) As described earlier, finding a workable solution that balances the amount of protective covering for the composite airfoil as described herein whilst maintaining a weight requirement is a labor-intensive and time-intensive process, because the process is iterative and involves the selection of multiple composite airfoils with various protector edge lengths and chord lengths.

Design procedures require placing said composite airfoil **130** (FIG. **2**) into a turbine engine designed for a first flight operating condition and embodying a protection effectiveness with acceptable weight gain/losses for that first flight operating condition. Evaluating whether in a second, third, or other flight operating condition, the same selected composite airfoil **130** maintains a heat effectiveness with acceptable protection effectiveness for the other operating conditions is time-intensive and necessitates re-design of the composite airfoil and even the turbine engine in the event the conditions are not met. It is desirable to have an ability to arrive at an optimal composite airfoil, like the composite airfoil(s) described herein, rather than relying on chance. It would be desirable to have a limited or narrowed range of possible composite airfoil configurations for satisfying mission requirements, such requirements including protection, weight restrictions, heat transfer, pressure ratio, and noise transmission level requirements, as well as the ability to survive bird strikes at the time a composite airfoil **130** is selected and located within an engine.
(98) The inventor(s) sought to find the trade-off balance between leading edge protection and weight gain/loss while satisfying all design requirements, because this would yield a more desired composite airfoil suited for specific needs of the engine, as described above. Knowing these trade-offs is also a desirable time saver.
(99) TABLE 1 below illustrates some composite airfoil configurations that yielded workable solutions to the trade-off balance problem.
(100) TABLE-US-00001 TABLE 1 Example: 1 2 3 4 5 6 CL (cm) 47 11 29 60 9.7 13 LL (cm) 11 1.7 3.2 16 1.5 2.3 SL (%) 20 20 38 50 50 80
(101) It was discovered, unexpectedly, during the course of engine design and the time-consuming iterative process previously described, that a relationship exists between the ratio of the leading length LL to the chord length CL. It has been found that the optimal amount of protective covering of the composite airfoil lies within a specific range based on the leading length LL of the protective covering and the chord length CL of the composite airfoil.
(102) TABLE 2 below illustrates some consecutive composite airfoil stages with workable solutions to the trade-off balance problem. Different span percentages are shown in TABLE 2. It was found that the CL and LL should be taken for any position between 20% and 80%, inclusive of end points of the span length SL. The specific range of the span length was chosen because the airfoil may have different properties, profiles, etc. at its distal ends. In the non-limiting examples, the fan blade dimensions determine APF1 while the outlet guide vane dimensions determine APF2.
(103) TABLE-US-00002 TABLE 2 Outlet Guide Vane Fan Blade Outlet Guide Vane Span (%) CL (cm) LL (cm) Span (%) CL (cm) LL (cm) 20 46.9 11.2 20 31.4 3.18 24 48.3 11.6 26 30.6 3.18 28 50.5 13.6 32 30.0 3.18 32 52.4 14.2 38 29.3 3.18 36 54.5 14.6 44 28.7 3.18 40 56.5 15.0 50 28.1 3.18 44 58.2 15.3 56 27.5 3.18 48 59.4 15.5 62 26.9 3.18 52 60.1 15.7 68 26.6 3.18 56 60.6 15.6 74 26.7 3.18 60 61.0 15.7 80 27.4 3.18 64 61.5 15.5 68 61.9 15.4 72 65.0 15.4 76 63.2 15.5 80 64.4 15.7
(104) Moreover, utilizing this relationship, the inventors found that the number of suitable or feasible composite airfoil possibilities for placement in a turbine engine that are capable of meeting the design requirements could be greatly reduced, thereby facilitating a more rapid down-selection of composite airfoils to consider as an engine is being developed. Such benefit provides more insight to the requirements for a given engine, and to the requirements for particular composite airfoil locations within the engine, long before specific technologies, integration, or system requirements are developed fully. The discovered relationship also avoids or prevents late-stage redesign while also providing the composite airfoil with a required protection effectiveness within given weight parameters.
(105) More specifically, the inventors found that a relationship between the first expression of the APF, APF1, and the second expression of the APF, APF2, optimizes the protection amount for successive stages of airfoils. This relationship was an unexpected discovery during the course of engine design—i.e., designing multistage airfoil sections such as by way of non-limiting examples

fan sections, fan blades, and outlet guide vanes and evaluating the impact that an amount of protection on the fan blade has on a needed amount of protection on the outlet guide vane, or vice versa. Narrowing the options down based on surrounding stages of airfoils can significantly decrease both material and time costs.

(106) In other words, an amount of protection provided by the first leading edge protector **240***a* on the first airfoil **230***a* can affect an amount of protection necessary for the second airfoil **230***b* downstream of the first airfoil **230***a*. This relationship between the multistage airfoils or successive airfoils, such as **230***a* and **230***b*, can be described by a stage performance factor (denoted "SPF") determined from a relationship between the APF1 and the APF2. The stage performance factor can generally be represented by a ratio of the first airfoil protection factor APF1 to the second airfoil protection factor APF2 represented by:

(107) $SPF = \frac{APF1}{APF2}$ (3)

(108) More specifically, it was found that for any position between 20% and 80%, inclusive of end points of the span length SL, a desired SPF value is greater than or equal to 0.70 and less than or equal to 4 ($0.7 \leq SPF \leq 4$). The specific range of the span length was chosen because the airfoil may have different properties, profiles, etc. at its distal ends. Conversely, at any position between 20% and 80%, inclusive of end points the airfoil is more uniform and therefore the determined ratios are applicable. It will be understood that because of its position and movement, the rotating fan blade will likely require more coverage from the leading edge protector as compared to a static airfoil or OGV, which is driving the relationship ratio to the 0.7 to 4.0 range. This is due to the fact that the rotating blade has a higher kinetic energy from impact and is driven by the rotating velocity of the airfoil.

(109) Utilizing this relationship, the inventors were able to arrive at a better performing airfoil in terms of protection amount with acceptable weight increase. The inventors found that the SPF for a set first set of airfoils and a second set of airfoils downstream from the first set of airfoils could be narrowed to an SPF range of greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$). Narrowing the SPF range provides more insight to the requirements for a given engine well before specific technologies, integration and system requirements are developed fully. For example, as the fan speed is reduced, coverage on the first leading edge **235***a* by the first leading edge protector can decrease such that the APF1 also decreases. Further, knowing a range for the SPF can prevent or minimize late-stage redesign, decrease material cost, and save time.

(110) The SPF value represents how an amount of protection on a first stage of airfoils, like the first stage of airfoils **200***a*, impacts an amount of protection necessary for any downstream airfoil stages with respect to the first set of airfoil stages.

(111) In one example, the set of fan blades **142** illustrated in FIG. **4** can have dimensions of the Fan Blade at 20% from TABLE 2 and the set of outlet guide vanes **182** can have dimensions of the Outlet Guide Vane at 20% from TABLE 2. This results in an APF1 value of (11.2/46.9) or 0.24 and an APF2 value of (3.18/31.4) or 0.10. Using the SPF ratio, an SPF value of (0.24/0.10) or 2.40 is found.

(112) In another example, the set of fan blades **142** illustrated in FIG. **4** can have dimensions of the Fan Blade at 68% from TABLE 2 and the set of outlet guide vanes **182** can have dimensions of the Outlet Guide Vane at 68% from TABLE 2. This results in an APF1 value of (15.4/61.9) or 0.25 and an APF2 value of (3.18/26.6) or 0.12. Using the SPF ratio, an SPF value of (0.25/0.12) or 2.1 is found.

(113) Some lower and upper bound values for each design parameter for determining Expression (3) are provided below in TABLE 3:

(114) TABLE-US-00003 TABLE 3 Parameter Lower Bound Upper Bound SL (%) 20 80 20 80 First Airfoil FCL (cm) 24 32 56 77 FLL (cm) 6 8 13 19 Second Airfoil SCL (cm) 9.9 9.3 31 27 SLL (cm) 1.6 1.5 4 3.5

(115) It was found that first and second airfoil pairs with dimensions fitting in the ranges set out in TABLE 4 below fit into the composite airfoil dimensions previously described herein. These ranges enable a minimum weight gain for a compact and proficiently protected composite airfoils in succession.
(116) TABLE-US-00004 TABLE 4 Ratio Narrow Range Broad Range SPF 0.95-2.5 0.70-4.0 APF1 0.22-0.25 0.20-0.30 APF2 0.10-0.12 0.08-0.17
(117) Pairs of first and second airfoils, with the second airfoils downstream of the first airfoils within the ranges provided can be assembled to conform with any fan section, or other downstream stage relationship for blades/vanes and blades/blades. This can include any number of engine designs including ducted and unducted engines as well as a direct-drive configuration and an indirect-drive configuration such as a speed reduction device or a geared-drive configuration.
(118) For example, FIG. **5** illustrates a gas turbine engine **310** as a high-bypass turbofan jet engine, sometimes also referred to as a turbofan engine, which can include the set of composite airfoils or first and second stages of composite airfoils as described herein. The gas turbine engine **310** defines an axial direction A (extending parallel to a longitudinal centerline **312** provided for reference), a radial direction R, and a circumferential direction CD extending about the longitudinal centerline **312**. In general, the gas turbine engine **310** includes a fan section **314** and a turbomachine **316** disposed downstream from the fan section **314**.
(119) The exemplary turbomachine **316** depicted generally includes a substantially tubular outer casing **318** that defines an annular inlet **320**. The outer casing **318** encases, in serial flow relationship, a compressor section including a booster or low pressure (LP) compressor **322** and a high pressure (HP) compressor **324**, a combustion section **326**, a turbine section including a high pressure (HP) turbine **328** and a low pressure (LP) turbine **330**, and a jet exhaust nozzle section **332**. A high pressure (HP) shaft **334**, which may additionally or alternatively be a spool, drivingly connects the HP turbine **328** to the HP compressor **324**. A low pressure (LP) shaft **336**, which may additionally or alternatively be a spool, drivingly connects the LP turbine **330** to the LP compressor **322**. The compressor section, combustion section **326**, turbine section, and jet exhaust nozzle section **332** together define a working gas flow path **337**.
(120) In the illustrated example, and by way of non-limiting example, the fan section **314** includes a fan **338** having a plurality of fan blades **340** coupled to a disk **342** in a spaced apart manner. As depicted, the fan blades **340** extend outwardly from disk **342** generally along the radial direction R.
(121) Each fan blade **340** is rotatable relative to the disk **342** about a pitch axis P by virtue of the fan blades **340** being operatively coupled to a suitable pitch change mechanism **344** configured to collectively vary the pitch of the fan blades **340**, e.g., in unison. The gas turbine engine **310** further includes a speed reduction device in the form of a power gearbox **346**, and the fan blades **340**, disk **342**, and pitch change mechanism **344** are together rotatable about the longitudinal centerline **312** by LP shaft **336** across the power gearbox **346**. The power gearbox **346** includes a plurality of gears for adjusting a rotational speed of the fan **338** relative to a rotational speed of the LP shaft **336**, such that the fan **338** may rotate at a more efficient fan speed. It will be understood that any suitable speed reduction device configured to adjust the rotation of the fan **338** relative to the LP shaft **336** can be utilized and that a power gearbox is merely one example thereof.
(122) The disk **342** is covered by rotatable front hub **348** of the fan section **314**. The front hub **348** is also sometimes referred to as a spinner. The front hub **348** is aerodynamically contoured to promote an airflow through the plurality of fan blades **340**.
(123) Additionally, the exemplary fan section **314** includes an annular fan casing or outer nacelle **350** that circumferentially surrounds the fan **338**, circumferentially surrounds at least a portion of the turbomachine **316**, or a combination thereof. It should be appreciated that the nacelle **350** is supported relative to the turbomachine **316** by a plurality of outlet guide vanes **352**, which can be a second stage of airfoils in the non-limiting example. Moreover, a downstream section **354** of the nacelle **350** extends over an outer portion of the turbomachine **316** so as to define a bypass airflow

passage **356** therebetween.

(124) It will be understood that each fan blade of the plurality of fan blades **340** may form a composite airfoil and that the plurality of fan blades **340** can form a first stage of airfoils as described above. More specifically, each of the plurality of fan blades **340** can include a first leading edge protector **340***a*. It will be understood that the plurality of fan blades forming the first stage of airfoils are similar to the previously described airfoils **130** and **230***a* with it being understood that the description of like parts applies to the plurality of fan blades unless otherwise noted.

(125) Further still, it will be understood that each outlet guide vane of the plurality of outlet guide vanes **352** may form a composite airfoil. Further still, in the illustrated example, the plurality of outlet guide vanes **352** can form a second stage of airfoils as described above. More specifically, each of the plurality of outlet guide vanes **352** can include a second leading edge protector **352***a*. It will be understood that an outlet guide vane of the plurality of outlet guide vanes **352** forming the second stage of airfoils is similar to the previously described airfoils **130** and **230***b* with it being understood that the description of like parts applies to the outlet guide vane of the plurality of outlet guide vanes **352** unless otherwise noted.

(126) It will be understood that the plurality of fan blades **340** and the plurality of outlet guide vanes **352** are similar to the previously described first and second airfoil pairs with dimensions fitting in the ranges set out in TABLE 4 above.

(127) During operation of the gas turbine engine **310**, a volume of air **358** enters the gas turbine engine **310** through an associated inlet **360** of the nacelle **350** and fan section **314**. As the volume of air **358** passes across the fan blades **340**, a first portion of air **362** is directed or routed into the bypass airflow passage **356** and a second portion of air **364** as indicated by arrow **364** is directed or routed into the working gas flow path **337**, or more specifically into the LP compressor **322**. The ratio between the first portion of air **362** and the second portion of air **364** is commonly known as a bypass ratio. A pressure of the second portion of air **364** is then increased as it is routed through the HP compressor **324** and into the combustion section **326**, where it is mixed with fuel and burned to provide combustion gases **366**.

(128) The combustion gases **366** are routed through the HP turbine **328** where a portion of thermal and/or kinetic energy from the combustion gases **366** is extracted via sequential stages of HP turbine stator vanes **368** that are coupled to the outer casing **318** and HP turbine rotor blades **370** that are coupled to the HP shaft **334**, thus causing the HP shaft **334** to rotate, which supports operation of the HP compressor **324**. The combustion gases **366** are then routed through the LP turbine **330** where a second portion of thermal and kinetic energy is extracted from the combustion gases **366** via sequential stages of LP turbine stator vanes **372** that are coupled to the outer casing **318** and LP turbine rotor blades **374** that are coupled to the LP shaft **336**, thus causing the LP shaft **336** to rotate, which supports operation of the LP compressor **322**, rotation of the fan **338**, or a combination thereof.

(129) The combustion gases **366** are subsequently routed through the jet exhaust nozzle section **332** of the turbomachine **316** to provide propulsive thrust. Simultaneously, the pressure of the first portion of air **362** is substantially increased as the first portion of air **362** is routed through the bypass airflow passage **356** before it is exhausted from a fan nozzle exhaust section **376** of the gas turbine engine **310**, also providing propulsive thrust. The HP turbine **328**, the LP turbine **330**, and the jet exhaust nozzle section **332** at least partially define a hot gas path **378** for routing the combustion gases **366** through the turbomachine **316**.

(130) As previously described the stages of airfoils exemplary gas turbine engine **310** depicted in FIG. **5** is by way of example only, and that in other exemplary embodiments, the gas turbine engine **310** may have other configurations. For example, although the gas turbine engine **310** depicted is configured as a ducted gas turbine engine (i.e., including the outer nacelle **350**, also referred to herein as a turbofan engine), in other embodiments, the gas turbine engine **310** may be an unducted

gas turbine engine (such that the fan **338** is an unducted fan, and the outlet guide vanes **352** are cantilevered from the outer casing **318**; see, e.g., FIG. **6**; also referred to herein as an open rotor engine). Additionally, or alternatively, although the gas turbine engine **310** depicted is configured as a variable pitch gas turbine engine (i.e., including a fan **338** configured as a variable pitch fan), in other embodiments, the gas turbine engine **310** may alternatively be configured as a fixed pitch gas turbine engine (such that the fan **338** includes fan blades **340** that are not rotatable about a pitch axis P).

(131) FIG. **6**, illustrates another non-limiting example of a gas turbine engine **400**, which can include the set of composite airfoils or first and second stages of composite airfoils as described herein. The exemplary gas turbine engine **400** of FIG. **4** may be configured in substantially the same manner as the exemplary gas turbine engine **310** described above with reference to FIG. **5**.

(132) For example, the exemplary gas turbine engine **400** defines an axial direction A, a radial direction R, and a circumferential direction CD. Moreover, the engine **400** defines an axial centerline, longitudinal axis or engine centerline **412** that extends along the axial direction A. In general, the axial direction A extends parallel to the engine centerline **412**, the radial direction R extends outward from and inward to the engine centerline **412** in a direction orthogonal to the axial direction A, and the circumferential direction CD extends three hundred sixty degrees (360°) around the engine centerline **412**. The engine **400** extends between a forward end **414** and an aft end **416**, e.g., along the axial direction A.

(133) Further, the exemplary gas turbine engine **400** generally includes a fan section **450** and a turbomachine **420**. Generally, the turbomachine **420** includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. In a non-limiting example, the turbomachine **420** includes a core cowl **422** that defines a core inlet **424** that is annular. The core cowl **422** further encloses at least in part a low pressure system and a high pressure system. For example, the core cowl **422** depicted encloses and supports at least in part a booster or low pressure (“LP”) compressor **426**, a high pressure (“HP”) compressor **428**, a combustor **430**, a high pressure turbine **432**, and a low pressure turbine **434**. The high pressure turbine **432** drives the high pressure compressor **428** through a high pressure shaft **436**. The low pressure turbine **434** drives the low pressure compressor **426** and components of the fan section **450** through a low pressure shaft **438**. After driving each of the high pressure turbine **432** and the low pressure turbine **434**, combustion products exit the turbomachine **420** through a turbomachine exhaust nozzle **440**.

(134) In this manner, the turbomachine **420** defines a working gas flow path or core duct **442** that extends between the core inlet **424** and the turbomachine exhaust nozzle **440**. The core duct **442** is an annular duct positioned generally inward of the core cowl **422** along the radial direction R. The core duct **442** may be referred to as a second stream.

(135) The fan section **450** includes a fan **452**, which is the primary fan in non-limiting example. One difference is that the fan **452** is an open rotor or unducted fan. In such a manner, the gas turbine engine **400** may be referred to as an open rotor engine. The fan **452** includes fan blades **454**, while only a single flan blade is illustrated in FIG. **6** it will be understood that an array of fan blades are included. Moreover, the fan blades **454** can be arranged in equal spacing around the engine centerline **412**. Each fan blade **454** has a root and a tip and a span defined therebetween. Each fan blade **454** defines a central blade axis **456**. For this embodiment, each fan blade **454** of the fan **452** is rotatable about its central blade axis **456**, e.g., in unison with one another. One or more actuators **458** are provided to facilitate such rotation and therefore may be used to change a pitch of the fan blades **454**.

(136) The fan blades **454** are rotatable about the engine centerline **412**. As noted above, the fan **452** is drivingly coupled with the low pressure turbine **434** via the LP shaft **438**. In a non-limiting example, the fan **452** is coupled with the LP shaft **438** via a speed reduction device, which can include by way of non-limiting examples a power gearbox or a speed reduction gearbox **455**, e.g., in an indirect-drive or geared-drive configuration.

(137) The fan section **450** further includes a fan guide vane array **460** that includes fan guide vanes **462**, again while only one fan guide vane is shown in FIG. **6** it will be understood that the fan guide vanes **462** are disposed around the engine centerline **412**. The fan guide vanes **462** are mounted to the fan cowl **470**. In a non-limiting example, the fan guide vanes **462** are not rotatable about the engine centerline **412**. Each of the fan guide vanes **462** has a root and a tip and a span defined therebetween. The fan guide vanes **462** may be unshrouded as shown in FIG. **6** or, alternatively, may be shrouded, e.g., by an annular shroud spaced outward from the tips of the fan guide vanes **462** along the radial direction R or attached to the fan guide vanes **462**.
(138) Each fan guide vane **462** defines a central blade axis **464**. By way of non-limiting example, each of the fan guide vanes **462** of the fan guide vane array **460** is rotatable about its respective central blade axis **464**, e.g., in unison with one another. One or more actuators **466** are provided to facilitate such rotation and therefore may be used to change a pitch of the fan guide vane **462** about its respective central blade axis **464**. However, in other embodiments, each of the fan guide vanes **462** may be fixed or unable to be pitched about its central blade axis **464**.
(139) It will be understood that each of the fan blades **454** may form a composite airfoil and that the fan blades **454** can form a first stage of airfoils as described above. More specifically, each of the fan blades **454** can include a first leading edge protector **454***a*. It will be understood that the fan blades **454** forming the first stage of airfoils are similar to the previously described airfoils **130**, **230***a*, and **340** with it being understood that the description of like parts applies to the fan blades unless otherwise noted.
(140) Further still, it will be understood that each of the fan guide vanes **462** may form a composite airfoil. Further still, in the illustrated example, the fan guide vanes **462** can form a second stage of airfoils as described above. More specifically, each of the fan guide vanes **462** can include a second leading edge protector **462***a*. It will be understood that the fan guide vanes **462** forming the second stage of airfoils is similar to the previously described airfoils **130**, **230***b*, and **352** with it being understood that the description of like parts applies to the fan guide vanes **462** unless otherwise noted.
(141) It will be understood that the fan blades **454** and the fan guide vanes **462** are similar to the previously described first and second airfoil pairs with dimensions fitting in the ranges set out in TABLE 4 above.
(142) Another difference is that the illustrated example in FIG. **6**, in addition to the unducted fan **452**, shows a ducted fan **484** included aft of the fan **452**. In this manner, the engine **400** includes both a ducted fan **484** and an unducted fan **452**, which both serve to generate thrust through the movement of air without passage through at least a portion of the turbomachine **420** (e.g., without passage through the HP compressor **428** and combustion section for the embodiment depicted). The ducted fan **484** is rotatable about the engine centerline **412**. The ducted fan **484** is, by way of non-limiting example, driven by the low pressure turbine **434** (e.g. coupled to the LP shaft **438**). The fan **452** may be referred to as the primary fan, and the ducted fan **484** may be referred to as a secondary fan. It will be appreciated that these terms “primary” and “secondary” are terms of convenience, and do not imply any particular importance, power, or the like.
(143) The ducted fan **484** includes a plurality of fan blades (not separately labeled in FIG. **6**) arranged in a single stage, such that the ducted fan **484** may be referred to as a single stage fan. The fan blades of the ducted fan **484** can be arranged in equal spacing around the engine centerline **412**. Each blade of the ducted fan **484** has a root and a tip and a span defined therebetween.
(144) The fan cowl **470** annularly encases at least a portion of the core cowl **422** and is generally positioned outward of at least a portion of the core cowl **422** along the radial direction R. Particularly, a downstream section of the fan cowl **470** extends over a forward portion of the core cowl **422** to define a fan duct flowpath, or simply a fan duct **472**. The fan flowpath or fan duct **472** may be understood as forming at least a portion of the third stream of the engine **400**.
(145) Incoming air may enter through the fan duct **472** through a fan duct inlet **476** and may exit

through a fan exhaust nozzle **478** to produce propulsive thrust. The fan duct **472** is an annular duct positioned generally outward of the core duct **442** along the radial direction R. The fan cowl **470** and the core cowl **422** are connected together and supported by a plurality of substantially radially extending, circumferentially-spaced stationary struts **474** (only one of which is shown in FIG. **6**). The stationary struts **474** may each be aerodynamically contoured to direct air flowing thereby. Other struts in addition to the stationary struts **474** may be used to connect and support the fan cowl **470**, the core cowl **422**, or a combination thereof. In many embodiments, the fan duct **472** and the core duct **442** may at least partially co-extend axially on opposite radial sides of the core cowl **422**. For example, the fan duct **472** and the core duct **442** may each extend directly from a leading edge **444** of the core cowl **422** and may partially co-extend generally axially on opposite radial sides of the core cowl **422**.

(146) The engine **400** also defines or includes an inlet duct **480**. The inlet duct **480** extends between the engine inlet **482** and the core inlet **424**, the fan duct inlet **476**, or a combination thereof. The engine inlet **482** is defined generally at the forward end of the fan cowl **470** and is positioned between the fan **452** and the fan guide vane array **460** along the axial direction A. The inlet duct **480** is an annular duct that is positioned inward of the fan cowl **470** along the radial direction R. Air flowing downstream along the inlet duct **480** is split, not necessarily evenly, into the core duct **442** and the fan duct **472** by a fan duct splitter or leading edge **444** of the core cowl **422**. In the embodiment depicted, the inlet duct **480** is wider than the core duct **442** along the radial direction R. The inlet duct **480** is also wider than the fan duct **472** along the radial direction R.

(147) Air passing through the fan duct **472** may be relatively cooler than one or more fluids utilized in the turbomachine **420**. In this way, one or more heat exchangers **486** may be positioned in thermal communication with the fan duct **472**. For example, one or more heat exchangers **486** may be disposed within the fan duct **472** and utilized to cool one or more fluids from the core engine with the air passing through the fan duct **472**, as a resource for removing heat from a fluid, e.g., compressor bleed air, oil or fuel. The heat exchanger **486** may be an annular heat exchanger.

(148) In various embodiments, it will be appreciated that the engine includes a ratio of a quantity of vanes to a quantity of blades that could be less than, equal to, or greater than 1:1. For example, in particular embodiments, the engine includes twelve (12) fan blades and ten (10) vanes. In other embodiments, the vane assembly includes a greater quantity of vanes to fan blades. For example, in particular embodiments, the engine includes ten (10) fan blades and twenty-three (23) vanes. For example, in certain embodiments, the engine may include a ratio of a quantity of vanes to a quantity of blades between 1:2 and 5:2. The ratio may be tuned based on a variety of factors including a size of the vanes to ensure a desired amount of swirl is removed for an airflow from the primary fan.

(149) It should be appreciated that various embodiments of the engine, such as the single unducted rotor engine depicted and described herein, may allow for normal subsonic aircraft cruise altitude operation at or above Mach 0.5. In certain embodiments, the engine allows for normal aircraft operation between Mach 0.55 and Mach 0.85 at cruise altitude. In still particular embodiments, the engine allows for normal aircraft operation between Mach 0.75 and Mach 0.85. In certain embodiments, the engine allows for rotor blade tip speeds at or less than 750 feet per second (fps). In other embodiments, the rotor blade tip speed at a cruise flight condition can be 650 to 900 fps, or 700 to 800 fps.

(150) It will be understood that a speed reduction device including, but not limited to, a gear assembly may be provided to reduce a rotational speed of the fan assembly relative to a driving shaft (such as a low pressure shaft coupled to a low pressure turbine). In some embodiments, a gear ratio of the input rotational speed to the output rotational speed is greater than or equal to 2. For example, in particular embodiments, the gear ratio is within a range of 4.1 to 14.0, within a range of 4.5 to 14.0, or within a range of 6.0 to 14.0. In certain embodiments, the gear ratio is within a range of 4.5 to 12 or within a range of 6.0 to 11.0. As such, in some embodiments, the fan can be

configured to rotate at a rotational speed of 700 to 1500 revolutions per minute (rpm) at a cruise flight condition, while the power turbine (e.g., the low-pressure turbine) is configured to rotate at a rotational speed of 2,500 to 15,000 rpm at a cruise flight condition. In particular embodiments, the fan can be configured to rotate at a rotational speed of 850 to 1,350 rpm at a cruise flight condition, while the power turbine is configured to rotate at a rotational speed of 5,000 to 10,000 rpm at a cruise flight condition.

(151) With respect to a turbomachine of the gas turbine engine, the compressors and/or turbines can include various stage counts. As disclosed herein, the stage count includes the number of rotors or blade stages in a particular component (e.g., a compressor or turbine). For example, in some embodiments, a low pressure compressor may include 1 to 8 stages, a high-pressure compressor may include 8 to 15 stages, a high-pressure turbine may include 1 to 2 stages, and/or a low pressure turbine (LPT) may include 3 to 7 stages. In particular, the LPT may have 4 stages, or between 4 and 7 stages. For example, in certain embodiments, an engine may include a one stage low pressure compressor, an 11 stage high pressure compressor, a two stage high pressure turbine, and 4 stages, or between 4 and 7 stages for the LPT. As another example, an engine can include a three stage low-pressure compressor, a 10 stage high pressure compressor, a two stage high pressure turbine, and a 7 stage low pressure turbine.

(152) The SPF is useful for making trade-offs when determining an amount of protection on a first airfoil in relationship to an amount on an airfoil downstream of the first airfoil. For example, when there is a limited space available for a fan blade in a fan section, knowledge of those dimensions and the downstream airfoil dimensions enables determination of an acceptable cover with a leading edge protector length allowing for sufficient leading edge protection.

(153) Benefits associated with the SPF described herein include a quick assessment of design parameters in terms of composite airfoils in downstream relationship. Further, the SPF described herein enables a quick visualization of tradeoffs in terms of geometry that are bounded by the constraints imposed by the materials used, the available space in which the composite airfoils are located, the type of turbine engine or system enclosures and the configuration of surrounding components, or any other design constraint. The SPF enables the manufacturing of a high performing composite airfoil with peak performance with the factors available. While narrowing these multiple factors to a region of possibilities saves time, money, and resources, the largest benefit is at the system level, where the composite airfoils described herein enable improved system performance. Previously developed composite airfoils may peak in one area of performance by design, but lose efficiency or lifetime benefits in another area of performance. In other words, the stage performance factor enables the development and production of higher performing composite airfoils across multiple performance metrics within a given set of constraints.

(154) As discussed, the inventors have conducted multiple tests, simulations, and analyses to assess and develop leading edge protector characteristics related to the ingestion of foreign objects such as birds. If a large foreign object impacts a fan blade, the fan blade, or a portion thereof, could break off from the rotor disk. A detached fan blade could damage adjacent fan blades and create a large imbalance in the fan assembly. Furthermore, if not contained by a fan casing, a detached fan blade could cause considerable damage to the aircraft powered by the engine. The inventors have sought to reduce the blade damage during an ingestion event.

(155) In doing so, the inventors sought to provide discrete (e.g., non-integral) platforms for joining fan blades or vanes to a rotor, stator, or the like. These separate platforms must have suitable strength for accommodating both centrifugal loads and impact loads, such as those due to a bird strike, during operation. The inventors have addressed prior known platform deficiencies by providing platforms that include a structural body portion and an integrally-formed flowpath surface portion. In one aspect, the flowpath surface portion defines a pair of wings that extend laterally beyond the structural body portion. The wings are frangible so as to break off if an adjacent fan blade rotates in response to an ingestion event throughout its rotation capability.

Although the wings are crushed during an ingestion event, the structural body portion, which provides the bulk of the platform's mass, stays relatively intact. Thus, very little of the platform's mass is lost so that most of the function of defining an inner flowpath boundary is retained. In another aspect, the inventors have implemented a platform having a wedge-shaped platform bumper for reducing stresses on a blade or vane.

(156) In this regard, as shown in FIGS. **1**, **7** and **8**, a fan **20** may include a plurality of discrete platforms **534** that secure fan blades **42** to a spool **50** (also referred to as a rotor disk **516**). A platform **534** has a radially outer surface **536** that extends between the respective adjacent fan blades **42** so as to collectively define an inner flowpath boundary for channeling air **79** between the fan blades **42**.

(157) Referring to FIGS. **7-9**, a single fan platform **534** is shown in greater detail. The platform **534** includes a unitary, integrally formed member **535** comprising a structural body portion **540** and a flowpath surface portion **542** which are joined in a substantially T-shaped configuration in cross-section (see FIG. **9**). As best seen in FIG. **7**, the platform **534** has a forward end **544** disposed near the disk forward side **520**, and an axially opposite aft end **546** disposed near the disk aft side **522**. The body portion **540** contributes the bulk of the platform's mass and consequently provides the platform **534** with the necessary strength to carry its centrifugal load.

(158) To reduce the overall weight of the platform **534** while maintaining suitable strength thereof, the integral member **535** is preferably made from a non-metal, composite material. For example, one suitable composite material would be graphite fibers embedded in an epoxy resin. Furthermore, the structural body portion **540** is a hollow body having a first side wall **550**, a second side wall **552** and a radially inner surface **554** extending between the two side walls **550**, **552**. The structural body portion **540** is open at both the forward end **544** and the aft end **546**. The open ends inhibit accumulation of fluids, such as rain water or melted ice, inside the hollow body portion **540** by allowing a centrifugal drainage path during engine operation. To further reduce weight, a number of weight relief holes **548**, shown in first side wall **550** in FIG. **7**, are formed in each side wall **550**, **552** of the structural body portion **540**.

(159) Referring to FIG. **8**, it is seen that the structural body portion **540** has a curved contour extending from the forward end **544** to the aft end **546**, as opposed to a straight wall configuration. This contour is chosen to substantially match the contour of the adjacent fan blades **42** between which the platform **534** will be disposed. That is, the first side wall **550** of the body portion **540** has a convex curvature that closely follows the concave curvature of pressure side of its adjacent blade, and the second side wall **552** of the body portion **540** has a concave curvature that closely follows the convex curvature of suction side of the blade adjacent to it.

(160) The structural body portion **540** also has a forward positioning bumper **556** attached to the radially inner surface **554** adjacent to the forward end **544** and an aft positioning bumper **558** attached to the radially inner surface **554** adjacent to the aft end **546**. Both the forward and aft positioning bumpers **556**, **558** are made of any suitable resilient material such as rubber and function to radially locate the platform **534** with respect to the rotor disk **516**, as will be described more fully below. The positioning bumpers **556**, **558** are secondarily bonded to the radially inner surface **554** of the body portion **540** in any suitable manner such as with adhesive.

(161) The flowpath surface portion **542** is integrally formed at the radially outermost extent of the structural body portion **540** so as to define the radially outer surface **536** of the platform **534** that fixes the inner flowpath boundary. As best seen in FIG. **9**, the flowpath surface portion **542** extends laterally (i.e., circumferentially) beyond each side wall **550**, **552** of the structural body portion **540** to define a pair of thin wings **560**. The wings **560** extend blade-to-blade so as to completely fill the space between adjacent blades **42**, thereby maintaining the inner flowpath boundary between the spinner **4** and the booster **4**. Like the side walls **550**, **552** of the body portion **540**, the outer lateral edges of the wings **560** are provided with a curved contour that matches the contour of the corresponding adjacent fan blade **42**. Furthermore, the outer lateral edge of each wing **560** is

provided with a resilient seal member **562** to seal fan blade air leakage during engine operation. The edge seals **562** are secondarily bonded to the wings **560**, preferably with a film adhesive, and are made of a suitable material such as silicone.

(162) The flowpath surface portion **542** provides the necessary strength to meet fan overspeed requirements, low cycle fatigue, and ingestion requirements, but the wings **560** are sufficiently thin so as to be frangible in the event of hard impact crush loads between a fan blade **42** and the platform **534** that can occur during ingestion events. As shown in FIG. **9**, the wings **560** are provided with a thickness, t, and a wing overhang or width, L. The ratio of the wing width, L, to the thickness, t, varies across the wings **560** because of their curved contour. Preferably, the aft portion of the wing **560** that is on the concave side of the platform **534** has a width-to-thickness ratio that is equal to or greater than 540 to provide sufficient frangibility. The width-to-thickness ratio varies between 520 and 540 for the other portions of the wings **560**.

(163) In addition, a thin glass fabric layer **543** is disposed on the radially outermost surface of the flowpath surface portion **542** for erosion protection. The glass fabric layer **543** has good erosion resistance and also serves a sacrificial function in the event of a foreign object impact. Specifically, as long as the platform's composite material is not penetrated as the result of a foreign object impact (i.e., only the glass fabric layer is damaged), then platform repair is typically a simple task. Penetration of the composite material will require a more difficult repair or replacement of the platform. The integral member **535** is provided with one or more coats of a high gloss polyurethane paint for additional erosion protection.

(164) Referring again to FIG. **7**, it can be seen that the platform **534** has a forward mounting flange **564** extending axially outward from the forward end **544** and an aft mounting flange **566** extending axially outward from the aft end **546**. The forward and aft mounting flanges **564**, **566** are configured so as to define forward and aft radially outward-facing abutment surfaces **568** and **570**, respectively, and forward and aft axially facing abutment surfaces **572** and **574**, respectively. Each abutment surface **568**, **570** has a wear strip, made of a wear resistant material such as an aramid fiber combined with polytetrafluoroethlene fiber woven into a suitable fabric, bonded thereto.

(165) The platform **534** is retained by a forward support ring **576** and an aft support ring **578**. The forward support ring **576** is an annular member that is substantially C-shaped in cross-section and includes a radially inner segment **580**, a radially outer segment **582**, and a middle portion **584** joining the two segments **580**, **582**. The inner segment **580** is fixedly joined at its distal end to the forward side **520** of the rotor disk **516** by a plurality of bolts, for example. The radially outer segment **582** overlaps the forward mounting flange **564** and engages the forward radial abutment surface **568**, thereby retaining the forward end **544** of the platform **534** against radially outward movement due to centrifugal force upon rotation of the rotor disk **516** during engine operation. Furthermore, the distal end of the outer segment **582** abuts the forward axially facing abutment surface **572** to restrain the platform **534** against forward axial movement. The middle portion **584** of the forward support ring **576** abuts the aft end of the spinner. Optionally, the forward support ring **576** may be an integral portion of the otherwise conventional spinner.

(166) The aft support ring **578** is an annular member that is substantially V-shaped in cross-section and includes a radially inner segment **586** and a radially outer segment **588** joined together at an intersection that defines an abutment **590**. The inner segment **586** is fixedly joined at its distal end to a mounting flange **592** formed on the booster shaft **528**. The abutment **590** overlaps the aft mounting flange **566** and engages the aft radial abutment surface **570**, thereby retaining the aft end **546** of the platform **534** against radially outward movement due to centrifugal force upon rotation of the rotor disk **516** during engine operation. The abutment **590** also engages the aft axially facing abutment surface **574** so as restrain the platform **534** against axial movement in the aft direction.

(167) During installation of the platform **534**, the forward and aft positioning bumpers **556**, **558** radially locate the platform **534** with respect to the rotor disk **516** to provide the desired clearance between the platform **534** and the radially outer surface of the rotor disk **516**. The forward

positioning bumper **556** contacts the inner segment **80** of the forward support ring **576**, and the aft positioning bumper **558** contacts the booster shaft **528**. The forward and aft positioning bumpers **556**, **558** also prevent the platform **534** from clanking against the rotor disk **516**, the booster shaft **528**, or the forward and aft support rings **576**, **578**.

(168) Fan blades **42** may be secured to the rotor disk **516** be known means. For example, a rotor disk **516** may include a plurality of circumferentially spaced apart axial dovetail slots which extend radially inwardly from the disk outer surface. Each of the fan blades **42** may include an integral root section in the form of a complementary axial-entry dovetail. The dovetail root sections are disposed in respective ones of the dovetail slots for attaching the fan blades **42** to the rotor disk **516**. As is known in the art, the dovetail slots and root sections are designed so as to permit limited rotation of the root section within the dovetail slot in response to an extreme force exerted on the blade **42**. This blade rotation capability substantially reduces the blade's susceptibility to foreign object damage.

(169) As discussed above, the body portion **540** has a curved contour that follows the contour of the adjacent fan blades **42**. The contoured body portion **540** allows increased blade rotation relative to the blade rotation possible with conventional fan platforms having a straight wall configuration by eliminating hard body pinch points between the platform **534**, the fan blades **42**, and the rotor disk **516** during ingestion events. The contoured configuration also facilitates installation of the platform **34**. The structural body portion **540** is sized and configured to provide adequate clearance with the rotor disk **516** and the adjacent blades **42** such that blade rotation capability is not overly hindered by binding between the platform **534** and the blades **42** or rotor disk **516**. Thus, if one of the fan blades **42** is struck by a foreign object, the blade **42** will rotate within its dovetail slot in response to the impact. As the blade **42** rotates, the frangible wings **560** will break off, allowing the blade **42** to rotate throughout its rotation capability, which is typically about 18 degrees. Although the wings **560** are crushed during an ingestion event, the structural body portion **540**, which provides the bulk of the platform's mass, stays relatively intact. Thus, very little of the platform's mass is lost so that most of the function of defining an inner flowpath boundary is retained.

(170) The platform **534** is preferably manufactured using a resin transfer molding (RTM) process in which fibers of a suitable material such as graphite are arranged in a desired orientation on a mandrel so as to approximate the desired shape of the finished part. This preform is then placed in a mold having a cavity matching the shape of the integral member **535**. A thin glass fabric mat is added to the mold, next to the surface of the preform that will become the flowpath surface portion **542** of the integral member **535**. The next step is to inject a resin such as an epoxy into the mold under moderate pressure so as to impregnate the fibers. The resin impregnated preform is then cured; the glass fabric mat is co-cured to form the glass fabric layer **543**. After curing, the mandrel is removed, and the integral member **535** is provided with one or more coats of a high gloss polyurethane paint.

(171) Once the integral member **535** is finished, the forward and aft positioning bumpers **556**, **558** are bonded to the radially inner surface **554** of the body portion **540**. The seal members **562** are bonded to the outer lateral edge of each wing **560**, and a wear strip is bonded to the forward and aft radially outward-facing abutment surfaces **568**, **570** of the forward and aft mounting flanges **564**, **566**, respectively. Thus, this is a relatively simple and inexpensive process for producing the lightweight fan platform of the present disclosure. Referring to FIG. **10**, another fan assembly **600** having non-integral fan platforms **700** is shown. The assembly includes a mounting plate **610** securable to a rotor disk **612**, and fan blades **614** securable to the rotor disk.

(172) Each of the fan blades **614** has a curved airfoil section **620** with pressure and suction sides **622** and **624**, respectively, extending between airfoil leading and trailing edges LE and TE, respectively. The airfoil section **620** is attached to a circular arc dovetail root **630** and a transition section **632** of the fan blade **614** extends between the airfoil section **620** and the root **630**.

(173) Referring to FIG. **11**, the fan rotor disk **612** is a multi-bore disk having a rim **640** attached to

a number of disk hubs **642** with bores **644** by a corresponding number of webs **646** circumscribed about a centerline of the rotor disk **612**. Web channels **648** extend axially between the webs **646** and radially between the rim **640** and the hubs **642**. Each of the disk posts **652** has an overhang **670** extending axially forwardly from the disk rim **640** and are located radially outwardly on the disk posts **652**. In the exemplary embodiment, the disk outer surface **660** of the rim **640** is contiguous with the disk posts **652** and the overhangs **670**.

(174) Referring again to FIG. **10**, a plurality of circumferentially spaced apart circular arc dovetail slots **650** are disposed through the rim **640** and extend circumferentially between disk posts **652**, axially from a forward end **654** to an aft end **656** of the rim, and radially inwardly from the disk outer surface **660** of the rim. The circular arc dovetail slots **650** are used for receiving and radially retaining the circular arc dovetail roots **630**.

(175) The circular arc dovetail root **630**, the circular arc dovetail slots **650**, and the disk posts **652** are arcuate and curved normal to and about a radial axis RA. This is exemplified by an arc AR through the disk post **652** is circumscribed about the radial axis RA at a radius of curvature R. Each of the circular arc dovetail roots **630** is designed to slide axially aftwardly along an arc into a corresponding one of a plurality circular arc dovetail slot **650** and be retained radially and circumferentially by the disk rim **640** and, more particularly, by the disk posts **652**. Though the sliding motion is circular along an arc, it is also referred to herein as an axially sliding motion.

(176) The platforms **700** have an aerodynamically contoured platform wall **702** with a radially outer surface **704** that faces radially outwardly and defines and maintains an inner fan flowpath that extends axially across the fan blade **614**. A radially inner surface **706** of the platform wall **702** faces radially inwardly. The platform walls **702** are sloped with respect to a centerline of the rotor disk **612** to provide an increasing radius of the outer surface **702** (the inner fan flowpath surface along the platform) in the axially aft direction.

(177) Referring to FIG. **12**, a wedge-shaped platform bumper **710** depends radially inwardly from the inner surface **706** of the platform wall **702**. The platform bumper **710**, in the exemplary embodiment, has a flat bottom surface **712**. The platform bumper **710**, in an alternative embodiment, may have a circumferentially curved bottom surface contoured to match the circumferentially curved disk outer surface **660**.

(178) The platform walls **702** have a rectangularly shaped forward portion **720** and a circumferentially curved aft portion **722**. The circumferentially curved aft portion **722** is contoured to fit around the curved airfoil section **620** between the airfoil leading and trailing edges LE and TE, respectively. The circumferentially curved aft portion **722** has pressure and suction side edges **724** and **726**, respectively, which are shaped to conform to the pressure and suction sides **622** and **624**, respectively, of the airfoil section **620**.

(179) Each of the platforms **700** also has forward, mid, and aft mounting lugs **730**, **732**, and **734**, respectively, depending radially inwardly from the platform wall **702**. The forward and aft mounting lugs **730** and **734** are located at forward and aft ends of the platform walls **702** and the mid mounting lug **732** is axially located therebetween, though, not necessarily midway.

(180) The rectangularly shaped forward portion **720** of the platform wall **702** includes a platform leading edge **736** extending axially forward just past the rim **640** and the forward mounting lug **730** depending from the forward portion **720**.

(181) The platforms **700** include circumferentially-curved aft stiffening ribs **740** extend between the mid and aft mounting lugs **732** and **734**. The aft stiffening ribs **740** extend substantially parallel to and spaced an inwardly from the pressure and suction side edges **722** and **724**, respectively. Circumferentially curved forward stiffening ribs **742** extend axially from the mid mounting lug **732** to a forward edge **744** of the platform bumper **710**, about where the wedge-shaped platform bumper **710** begins to depend radially inwardly from the inner surface **706** of the platform wall **702**. The forward stiffening ribs **742** are tapered or blended down to the inner surface **706** of the platform **700**, such that at any axial position, the height of the forward stiffening ribs is less than the height

of the platform bumper **710** along an axially extending bumper length. The platform bumper **710** provides additional stiffness to control the stress and deflection of the platform **700** and platform wall **702** during ice or bird impacts. The platform bumper **710** creates a load path from the thin platform wall **702** into the top of the disk post **652** and limits deflections (and thus stresses) in case of such an impact event.

(182) Each of the platforms **700** is mounted on the disk **612** between two adjacent ones of the fan blades **614**. First, two adjoining fan blades are mounted on the disk **612** by circularly sliding the dovetail roots **630** into the corresponding dovetail slots **650** until the transition section **632** of the fan blade **614** contacts the annular mounting plate **610**. Thus, the annular mounting plate **610** provides aftwardly axial retention of the fan blade **614**. A platform **700** is then mounted on the disk post **652** in between the two adjacent mounted fan blades **614** by sliding the platform **700** axially aftwardly.

(183) To the extent one or more structures provided herein can be known in the art, it should be appreciated that the present disclosure can include combinations of structures not previously known to combine, at least for reasons based in part on conflicting benefits versus losses, desired modes of operation, or other forms of teaching away in the art.

(184) This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

(185) Further aspects are provided by the subject matter of the following clauses:

(186) A turbine engine, comprising: an engine core defining an engine centerline and comprising a rotor and a stator, a first stage of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge, a first leading edge protector comprising a first sheath receiving the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL), and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL), a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge, a second leading edge protector comprising a second sheath receiving the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL), and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL), wherein the first leading length (FLL) and the first chord length (FCL) relate to the second leading length (SLL) and the second chord length (SCL) by an expression: $((FLL/FCL))/((SLL/SCL))$ to define a stage protection factor (SPF), and wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$).

(187) The turbine engine of any proceeding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils include a polymer matrix composite (PMC).

(188) The turbine engine of any proceeding clause, wherein at least one of the first leading edge protector or the second leading edge protector is a metallic leading edge protector.
(189) The turbine engine of any proceeding clause, wherein the first stage of composite airfoils are fan blades.
(190) The turbine engine of any proceeding clause, wherein the second stage of composite airfoils are outlet guide vanes.
(191) The turbine engine of any proceeding clause, wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length and wherein the second airfoil extends spanwise between a second root and a second tip to define a second span length.
(192) The turbine engine of any proceeding clause, wherein the SPF is determined between 20% and 80% of the first span length and the second span length, inclusive of endpoints.
(193) The turbine engine of any proceeding clause, wherein the first stage of composite airfoils has a first number of airfoils and the second stage of composite airfoils has a second number of airfoils and the first number is different than the second number.
(194) The turbine engine of any proceeding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils are configured to rotate.
(195) The turbine engine of any proceeding clause, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$).
(196) The turbine engine of any proceeding clause, wherein the first sheath and the second sheath each have a first wall, a second wall, and a third wall interconnecting the first wall and the second wall.
(197) The turbine engine of any proceeding clause, wherein the first wall, second wall, and third wall of the leading edge protector are oriented and shaped such that they define a U-shaped or C-shaped channel therebetween.
(198) The turbine engine of any proceeding clause, wherein the channel is sized and shaped to receive the composite leading edge of the composite portion.
(199) The turbine engine of any proceeding clause, wherein any of the first leading edge protector or the second leading edge protector are coupled to their corresponding composite portion at the corresponding composite leading edge to define at least one seam.
(200) The turbine engine of any proceeding clause wherein the at least one seam is two seams on either side of the airfoil, and the corresponding first leading length or second leading length is measured from the corresponding leading edge to the seam furthest from the leading edge.
(201) The turbine engine of any proceeding clause wherein the first leading length and the second leading length are measured from their corresponding leading edge to their corresponding seam.
(202) The turbine engine of any proceeding clause wherein an amount of overlap between the first sheath or the second sheath and their corresponding.
(203) A turbine engine comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a set of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the leading length (LL) and the chord length (CL) relate to each other by an expression: ((LL))/((CL)) to define an airfoil protection factor (APF); and wherein the APF is greater than or equal to 0.1 and less than or equal to 0.3 ($0.1 \leq APF \leq 0.3$).
(204) The turbine engine of any proceeding clause, wherein the set of composite airfoils includes a first stage of composite airfoils and a second stage of composite airfoils downstream from the first stage of composite airfoils.

(205) The turbine engine of any proceeding clause, wherein the first stage of composite airfoils is a set of fan blades and the second stage of composite airfoils is a set of outlet guide vanes.
(206) The turbine engine of any proceeding clause, wherein the first stage of composite airfoils has a first airfoil protection factor (APF1) and the second stage of composite airfoils has a second airfoil protection factor (APF2).
(207) The turbine engine of any proceeding clause, wherein the first airfoil protection factor (APF1) relates to the second airfoil protection factor (APF2) by an expression: APF1/APF2 to define a stage protection factor (SPF), wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$).
(208) The turbine engine of any proceeding clause, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$).
(209) The turbine engine of any proceeding clause, wherein the set of composite airfoils extend spanwise between a root and a tip to define a span length and the APF is determined between 20% and 80% of the first span length and the second span length.
(210) The turbine engine of any proceeding clause, wherein the leading edge protector overlaps with the composite leading edge to define a sheath.
(211) The turbine engine of any proceeding clause, wherein the composite portion is formed from a polymer matrix composite (PMC).
(212) The turbine engine of any proceeding clause, wherein the leading edge protector is a metallic leading edge protector.
(213) An indirect drive turbine engine, comprising: an engine core defining an engine centerline and comprising a rotor defined by a fan including a plurality of fan blades rotatable about the engine centerline, a stator, a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, a speed reduction device driven by the turbine section for rotating the fan about the engine centerline, a first stage of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge, a first leading edge protector comprising a first sheath receiving the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL), and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL), a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge, a second leading edge protector comprising a second sheath receiving the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL), and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL), wherein the first leading length (FLL) and the first chord length (FCL) relate to the second leading length (SLL) and the second chord length (SCL) by an expression: ((FLL/FCL))/((SLL/SCL)) to define a stage protection factor (SPF), and wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$).
(214) The indirect drive turbine engine of any proceeding clause, wherein the speed reduction device is a power gearbox.
(215) The indirect drive turbine engine of any proceeding clause, wherein the turbine section includes a fan drive turbine and a second turbine and the second turbine is disposed forward of the

fan drive turbine and the fan drive turbine includes a plurality of fan drive turbine stages with the speed reduction device driven by the fan drive turbine.
(216) The indirect drive turbine engine of any proceeding clause, further comprising a fan casing or nacelle.
(217) The indirect drive turbine engine of any proceeding clause, wherein the fan drive turbine has between 3 and 5 stages.
(218) The indirect drive turbine engine of any proceeding clause, wherein the speed reduction device is a power gearbox having a power gearbox reduction ratio between 2:1 and 5:1.
(219) The indirect drive turbine engine of any proceeding clause, wherein a bypass ratio is between 10:1 and 22:1.
(220) The indirect drive turbine engine of any proceeding clause, wherein a fan blade tip speed of the fan is less than 1400 feet per second.
(221) The indirect drive turbine engine of any proceeding clause, wherein the core is an open rotor engine.
(222) The indirect drive turbine engine of any proceeding clause, wherein the speed reduction device is a power gearbox having a power gearbox reduction ratio between 6:1 and 12:1.
(223) The indirect drive turbine engine of any proceeding clause, wherein a bypass ratio is between 25:1 and 125:1.
(224) The indirect drive turbine engine of any proceeding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils include a polymer matrix composite (PMC).
(225) The indirect drive turbine engine of any proceeding clause, wherein at least one of the first leading edge protector or the second leading edge protector is a metallic leading edge protector.
(226) The indirect drive turbine engine of any proceeding clause, wherein the first stage of composite airfoils include the plurality of fan blades.
(227) The indirect drive turbine engine of any proceeding clause, wherein the second stage of composite airfoils are outlet guide vanes or fan guide vanes.
(228) The indirect drive turbine engine of any proceeding clause, wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length and wherein the second airfoil extends spanwise between a second root and a second tip to define a second span length.
(229) The indirect drive turbine engine of any proceeding clause, wherein the SPF is determined between 20% and 80% of the first span length and the second span length, inclusive of endpoints.
(230) The indirect drive turbine engine of any proceeding clause, wherein the first stage of composite airfoils has a first number of airfoils and the second stage of composite airfoils has a second number of airfoils and the first number is different than the second number.
(231) The indirect drive turbine engine of any proceeding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils are configured to rotate.
(232) The indirect drive turbine engine of any proceeding clause, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$).
(233) The indirect drive turbine engine of any proceeding clause, wherein the first sheath and the second sheath each have a first wall, a second wall, and a third wall interconnecting the first wall and the second wall.
(234) The indirect drive turbine engine of any proceeding clause, wherein the first wall, second wall, and third wall of the leading edge protector are oriented and shaped such that they define a U-shaped or C-shaped channel therebetween.
(235) The indirect drive turbine engine of any proceeding clause, wherein the channel is sized and shaped to receive the composite leading edge of the composite portion.
(236) The indirect drive turbine engine of any proceeding clause, wherein any of the first leading edge protector or the second leading edge protector are coupled to their corresponding composite portion at the corresponding composite leading edge to define at least one seam.

(237) The indirect drive turbine engine of any proceeding clause wherein the at least one seam is two seams on either side of the airfoil, and the corresponding first leading length or second leading length is measured from the corresponding leading edge to the seam furthest from the leading edge.
(238) The indirect drive turbine engine of any proceeding clause wherein the first leading length and the second leading length are measured from their corresponding leading edge to their corresponding seam.
(239) The indirect drive turbine engine of any proceeding clause wherein an amount of overlap between the first sheath or the second sheath and their corresponding.
(240) An indirect drive turbine engine comprising: an engine core defining an engine centerline and comprising a rotor defined by a fan including a plurality of fan blades rotatable about the engine centerline, a stator, a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, a speed reduction device driven by the turbine section for rotating the fan about the engine centerline; a set of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the leading length (LL) and the chord length (CL) relate to each other by an expression: ((LL))/((CL)) to define an airfoil protection factor (APF); and wherein the APF is greater than or equal to 0.1 and less than or equal to 0.3 ($0.1 \leq APF \leq 0.3$).
(241) The indirect drive turbine engine of any proceeding clause, wherein the turbine section includes a fan drive turbine and a second turbine and the second turbine is disposed forward of the fan drive turbine and the fan drive turbine includes a plurality of fan drive turbine stages with the speed reduction device driven by the fan drive turbine.
(242) The indirect drive turbine engine of any proceeding clause, further comprising a fan casing or nacelle.
(243) The indirect drive turbine engine of any proceeding clause, wherein the fan drive turbine has between 3 and 5 stages.
(244) The indirect drive turbine engine of any proceeding clause, wherein the speed reduction device is a power gearbox having a power gearbox reduction ratio between 2:1 and 5:1.
(245) The indirect drive turbine engine of any proceeding clause, wherein a bypass ratio is between 10:1 and 22:1.
(246) The indirect drive turbine engine of any proceeding clause, wherein a fan blade tip speed of the fan is less than 1400 feet per second.
(247) The indirect drive turbine engine of any proceeding clause, wherein the core is an open rotor engine.
(248) The indirect drive turbine engine of any proceeding clause, wherein the speed reduction device is a power gearbox having a power gearbox reduction ratio between 6:1 and 12:1.
(249) The indirect drive turbine engine of any proceeding clause, wherein a bypass ratio is between 25:1 and 125:1.
(250) The indirect drive turbine engine of any proceeding clause, wherein the set of composite airfoils includes a first stage of composite airfoils and a second stage of composite airfoils downstream from the first stage of composite airfoils.
(251) The indirect drive turbine engine of any proceeding clause, wherein the first stage of composite airfoils is the plurality of fan blades and the second stage of composite airfoils is a set of outlet guide vanes or a set of fan guide vanes.
(252) The indirect drive turbine engine of any proceeding clause, wherein the first stage of

composite airfoils has a first airfoil protection factor (APF1) and the second stage of composite airfoils has a second airfoil protection factor (APF2).

(253) The indirect drive turbine engine of any proceeding clause, wherein the first airfoil protection factor (APF1) relates to the second airfoil protection factor (APF2) by an expression: APF1/APF2 to define a stage protection factor (SPF), wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$).

(254) The indirect drive turbine engine of any proceeding clause, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$).

(255) The indirect drive turbine engine of any proceeding clause, wherein the set of composite airfoils extend spanwise between a root and a tip to define a span length and the APF is determined between 20% and 80% of the first span length and the second span length.

(256) The indirect drive turbine engine of any proceeding clause, wherein the leading edge protector overlaps with the composite leading edge to define a sheath.

(257) The indirect drive turbine engine of any proceeding clause, wherein the composite portion is formed from a polymer matrix composite (PMC).

(258) The indirect drive turbine engine of any proceeding clause, wherein the leading edge protector is a metallic leading edge protector.

(259) A turbine engine, comprising: an engine core defining an engine centerline and comprising a rotor and a stator, a first stage of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge, a first leading edge protector comprising a first sheath receiving the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL), and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL), a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge, a second leading edge protector comprising a second sheath receiving the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL), and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL), wherein the first leading length (FLL) and the first chord length (FCL) relate to the second leading length (SLL) and the second chord length (SCL) by an expression: $((FLL/FCL))/((SLL/SCL))$ to define a stage protection factor (SPF), and wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$), wherein at least one of the first airfoil and the second airfoil is secured to the rotor or the stator by a platform, the platform providing a flowpath boundary, the platform comprising: a structural body portion comprising a hollow body having a first side wall, a second side wall, a radially inner surface extending between said first and second side walls, and open ends, the structural body portion having a contour that matches that of adjacent fan blades; and a flowpath surface portion joined to the structural body portion, the flowpath surface portion defining a pair of frangible wings that extend laterally beyond the structural body portion.

(260) A turbine engine comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a set of composite airfoils comprising a set of fan blades and a set of outlet guide vanes downstream from the set of fan blades, the set of composite airfoils circumferentially

arranged about the engine centerline and defining at least a portion of the rotor. An airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the leading length (LL) is related to the chord length (CL) by an airfoil protection factor

(261) $(APF) = \frac{(LL)}{(CL)}$

and a first APF (APF1) for the set of fan blades is greater than or equal to 0.2 and less than or equal to 0.30 ($0.2 \leq APF1 \leq 0.3$) and a second APF (APF2) for the set of outlet guide vanes is greater than or equal to 0.08 and less than or equal to 0.17 ($0.08 \leq APF2 \leq 0.17$); wherein at least one of a fan blade or and outlet guide vane is secured to the rotor or the stator by a platform, the platform providing a flowpath boundary, the platform comprising: a structural body portion comprising a hollow body having a first side wall, a second side wall, a radially inner surface extending between said first and second side walls, and open ends, the structural body portion having a contour that matches that of adjacent fan blades; and a flowpath surface portion joined to the structural body portion, the flowpath surface portion defining a pair of frangible wings that extend laterally beyond the structural body portion.

(262) The turbine engine of any proceeding clause, wherein the flowpath surface portion is integrally formed on the structural body portion to define a substantially T-shaped configuration.

(263) The turbine engine of any proceeding clause, wherein the structural body portion and the flowpath surface portion form an integral member that is made of a composite material that includes graphite fibers in an epoxy resin.

(264) The turbine engine of any proceeding clause, wherein the integral member is provided with at least one coat of polyurethane paint.

(265) The turbine engine of any proceeding clause, wherein each one of the first and second side walls has at least one weight relief hole formed therein.

(266) The turbine engine of any proceeding clause, wherein the first side wall has a concave curvature and the second side wall has a convex curvature.

(267) The turbine engine of any proceeding clause, further comprising a forward positioning bumper attached to the radially inner surface and an aft positioning bumper attached to the radially inner surface.

(268) The turbine engine of any proceeding clause, wherein the wings have a width and a thickness and the ratio of the width to the thickness is greater than or equal to 40 in at least a portion of the frangible wings.

(269) The turbine engine of any proceeding clause, further comprising a seal member attached to the frangible wings.

(270) The turbine engine of any proceeding clause, further comprising a glass fabric layer disposed on the flowpath surface portion.

(271) The turbine engine of any proceeding clause, further comprising a forward mounting flange extending axially outward from a forward end of the platform and an aft mounting flange extending axially outward from an aft end of the platform.

(272) A turbine engine, comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a first stage of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge; a first leading edge protector receiving at least a portion of the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading

length (FLL); and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL); a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge; a second leading edge protector receiving at least a portion of the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL); and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL); wherein the first leading length (FLL) and the first chord length (FCL) are related to the second leading length (SLL) and the second chord length (SCL) by a stage protection factor (SPF), wherein

(273) $SPF = \frac{(FLL / FCL)}{(SLL / SCL)}$

and SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$); wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length and the second airfoil extends spanwise between a second root and a second tip to define a second span length less than the first span length; wherein the turbine engine further includes a platform for circumferential disposition between fan blades, the platform comprising: a platform wall with a radially outer surface which faces radially outwardly and a radially inner surface which faces radially inwardly, the platform wall is sloped with respect to a centerline with an increasing radius of the outer surface in an axially aft direction, forward, mid, and aft mounting lugs depend radially inwardly from the platform wall, and a wedge-shaped platform bumper depends radially inwardly from the inner surface of the platform wall and is axially located forward of the mid mounting lug.

(274) A turbine engine comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a set of composite airfoils comprising a set of fan blades and a set of outlet guide vanes downstream from the set of fan blades, the set of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the leading length (LL) is related to the chord length (CL) by an airfoil protection factor

(275) $(APF) = \frac{(LL)}{(CL)}$

and a first APF (APF1) for the set of fan blades is greater than or equal to 0.2 and less than or equal to 0.30 ($0.2 \leq APF1 \leq 0.3$) and a second APF (APF2) for the set of outlet guide vanes is greater than or equal to 0.08 and less than or equal to 0.17 ($0.08 \leq APF2 \leq 0.17$); wherein the turbine engine further includes a platform for circumferential disposition between fan blades, the platform comprising: a platform wall with a radially outer surface which faces radially outwardly and a radially inner surface which faces radially inwardly, the platform wall is sloped with respect to a centerline with an increasing radius of the outer surface in an axially aft direction, forward, mid, and aft mounting lugs depend radially inwardly from the platform wall, and a wedge-shaped platform bumper depends radially inwardly from the inner surface of the platform wall and is axially located forward of the mid mounting lug.

(276) The turbine engine of any proceeding clause, further comprising circumferentially curved forward stiffening ribs extending axially from the mid mounting lug to a forward edge of the

platform bumper about where the platform bumper begins to depend radially inwardly from the inner surface of the platform wall.

(277) The turbine engine of any proceeding clause, wherein the forward stiffening ribs are tapered down to the inner surface of the platform such that at any axial location a first height of the forward stiffening ribs is less than a second height of the platform bumper along an axially extending bumper length.

(278) The turbine engine of any proceeding clause, further comprising circumferentially curved aft stiffening ribs extending between the mid and aft mounting lugs.

(279) The turbine engine of any proceeding clause, wherein the platform wall further comprises a rectangularly shaped forward portion and a circumferentially curved aft portion.

(280) The turbine engine of any proceeding clause, wherein the circumferentially curved aft portion has pressure and suction side edges that have pressure and suction side airfoil shapes respectively.

(281) The turbine engine of any proceeding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils include a polymer matrix composite (PMC).

(282) The turbine engine of any proceeding clause, wherein at least one of the first leading edge protector or the second leading edge protector is a metallic leading edge protector.

(283) The turbine engine of any proceeding clause, wherein the composite airfoils of the first stage of composite airfoils are fan blades and the composite airfoils of the second stage of composite airfoils are outlet guide vanes.

(284) The turbine engine of any proceeding clause, wherein the SPF is determined at a location between 20% and 80% of the first span length and the second span length, inclusive of endpoints.

(285) The turbine engine of any proceeding clause, wherein the first leading edge protector and the second leading edge protector each comprise a sheath.

(286) The turbine engine of any proceeding clause, wherein the first stage of composite airfoils has a first number of airfoils and the second stage of composite airfoils has a second number of airfoils and the first number is different than the second number.

(287) The turbine engine of any proceeding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils are configured to rotate.

(288) The turbine engine of any proceeding clause, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$).

(289) An indirect drive turbine engine, comprising: a fan; a turbomachine defining an engine centerline and comprising a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, and a speed reduction device driven by the turbine section for rotating the fan; the fan comprising a first stage of composite airfoils circumferentially arranged about the engine centerline, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge; a first leading edge protector comprising a first sheath receiving at least a portion of the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL); and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL); a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge; a second leading edge protector comprising a second sheath receiving at least a portion of the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL); and the second composite portion and the second leading edge protector together defining an exterior

surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL); wherein the first leading length (FLL) and the first chord length (FCL) relate to the second leading length (SLL) and the second chord length (SCL) by a stage protection factor (SPF); wherein

(290) $SPF = \frac{\left(\frac{FLL}{FCL}\right)}{\left(\frac{SLL}{SCL}\right)}$

and the SPF is greater than or equal to 0.7 and less than or equal to 4 (0.7≤SPF≤4); wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length and the second airfoil extends spanwise between a second root and a second tip to define a second span length less than the first span length; wherein at least one airfoil of the first and second stages of airfoils is secured to the turbomachine by a platform, the platform providing a flowpath boundary, the platform comprising: a structural body portion comprising a hollow body having a first side wall, a second side wall, a radially inner surface extending between said first and second side walls, and open ends, the structural body portion having a contour that matches that of adjacent fan blades; and a flowpath surface portion joined to the structural body portion, the flowpath surface portion defining a pair of frangible wings that extend laterally beyond the structural body portion.

(291) An indirect drive turbine engine comprising: a fan; a turbomachine defining an engine centerline and comprising a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, and a speed reduction device driven by the turbine section for rotating the fan; a set of composite airfoils comprising a set of fan blades and a set of outlet guide vanes downstream from the set of fan blades, the set of composite airfoils circumferentially arranged about the engine centerline, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the leading length (LL) and the chord length (CL) relate to each other by [an expression: (LL)/(CL) to define] an airfoil protection factor (APF),

(292) $(APF) = \frac{(LL)}{(CL)}$

and a first airfoil protection factor (APF1) for the set of fan blades is greater than or equal to 0.2 and less than or equal to 0.30 (0.2≤APF1≤0.3) and a second airfoil protection factor (APF2) for the set of outlet guide vanes is greater than or equal to 0.08 and less than or equal to 0.17 (0.08≤APF2≤0.17); wherein at least one of a fan blade of the set of fan blades and an outlet guide vane of the set of outlet guide vanes is secured to the turbomachine by a platform, the platform providing a flowpath boundary, the platform comprising: a structural body portion comprising a hollow body having a first side wall, a second side wall, a radially inner surface extending between said first and second side walls, and open ends, the structural body portion having a contour that matches that of adjacent fan blades; and a flowpath surface portion joined to the structural body portion, the flowpath surface portion defining a pair of frangible wings that extend laterally beyond the structural body portion.

(293) An indirect drive turbine engine, comprising: a fan; a turbomachine defining an engine centerline and comprising a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, and a speed reduction device driven by the turbine section for rotating the fan; the fan comprising a first stage of composite airfoils circumferentially arranged about the engine centerline, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge; a first leading edge protector comprising a first sheath receiving at least a portion of the first composite leading edge of the first composite

portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL); and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL); a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge; a second leading edge protector comprising a second sheath receiving at least a portion of the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL); and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL); wherein the first leading length (FLL) and the first chord length (FCL) relate to the second leading length (SLL) and the second chord length (SCL) by a stage protection factor (SPF); wherein
(294) $SPF = \frac{\left(\frac{FLL}{FCL}\right)}{\left(\frac{SLL}{SCL}\right)}$
and the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$); wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length and the second airfoil extends spanwise between a second root and a second tip to define a second span length less than the first span length; wherein the indirect drive turbine engine further includes a platform for circumferential disposition between fan blades, the platform comprising: a platform wall with a radially outer surface which faces radially outwardly and a radially inner surface which faces radially inwardly, the platform wall is sloped with respect to a centerline with an increasing radius of the outer surface in the axially aft direction, forward, mid, and aft mounting lugs depend radially inwardly from the platform wall, and a wedge-shaped platform bumper depends radially inwardly from the inner surface of the platform wall and is axially located forward of the mid mounting lug.
(295) An indirect drive turbine engine comprising: a fan; a turbomachine defining an engine centerline and comprising a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, and a speed reduction device driven by the turbine section for rotating the fan; a set of composite airfoils comprising a set of fan blades and a set of outlet guide vanes downstream from the set of fan blades, the set of composite airfoils circumferentially arranged about the engine centerline, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the leading length (LL) and the chord length (CL) relate to each other by [an expression: (LL)/(CL) to define] an airfoil protection factor (APF), (APF)=(LL)/(CL) and a first airfoil protection factor (APF1) for the set of fan blades is greater than or equal to 0.2 and less than or equal to 0.30 ($0.2 \leq APF1 \leq 0.3$) and a second airfoil protection factor (APF2) for the set of outlet guide vanes is greater than or equal to 0.08 and less than or equal to 0.17 ($0.08 \leq APF2 \leq 0.17$); wherein the indirect drive turbine engine further includes a platform for circumferential disposition between fan blades, the platform comprising: a platform wall with a radially outer surface which faces radially outwardly and a radially inner surface which faces radially inwardly, the platform wall is sloped with respect to a centerline with an increasing radius of the outer surface in an axially aft direction,

forward, mid, and aft mounting lugs depend radially inwardly from the platform wall, and a wedge-shaped platform bumper depends radially inwardly from the inner surface of the platform wall and is axially located forward of the mid mounting lug.

Claims

1. A turbine engine, comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a first stage of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge; a first leading edge protector receiving at least a portion of the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL); and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL); a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge; a second leading edge protector receiving at least a portion of the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL); and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL); wherein the FLL and the FCL are related to the SLL and the SCL by a stage protection factor (SPF), wherein the $SPF = \frac{(FLL / FCL)}{(SLL / SCL)}$, and the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$); wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length and the second airfoil extends spanwise between a second root and a second tip to define a second span length less than the first span length; wherein at least one of the first airfoil and the second airfoil is secured to the rotor or the stator by a platform, the platform providing a flowpath boundary, the platform comprising: a structural body portion comprising a hollow body having a first side wall, a second side wall, a radially inner surface extending between said first and second side walls, and open ends, the structural body portion having a contour that matches that of adjacent structural body portions; and a flowpath surface portion joined to the structural body portion, the flowpath surface portion defining a pair of frangible wings that extend laterally beyond the structural body portion.
2. The turbine engine of claim 1 wherein the turbine engine is an indirect turbine engine comprising a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, and a speed reduction device driven by the turbine section for rotating the fan.
3. The turbine engine of claim 1 wherein the flowpath surface portion is integrally formed on the structural body portion to define a substantially T-shaped configuration.
4. The turbine engine of claim 1 wherein the structural body portion and the flowpath surface portion form an integral member that is made of a composite material that includes graphite fibers in an epoxy resin.
5. The turbine engine of claim 4 wherein the integral member is provided with at least one coat of polyurethane paint.
6. The turbine engine of claim 1 wherein each one of the first and second side walls has at least one

weight relief hole formed therein.
7. The turbine engine of claim 1 wherein the first side wall has a concave curvature and the second side wall has a convex curvature.
8. The turbine engine of claim 1 further comprising a forward positioning bumper attached to the radially inner surface and an aft positioning bumper attached to the radially inner surface.
9. The turbine engine of claim 1 wherein the frangible wings have a width and a thickness and a ratio of the width to the thickness is greater than or equal to 40 in at least a portion of the frangible wings.
10. The turbine engine of claim 1 further comprising a seal member attached to the frangible wings.
11. The turbine engine of claim 1 further comprising a glass fabric layer disposed on the flowpath surface portion.
12. The turbine engine of claim 1 further comprising a forward mounting flange extending axially outward from a forward end of the platform and an aft mounting flange extending axially outward from an aft end of the platform.
13. The turbine engine of claim 1 wherein the first stage of composite airfoils and the second stage of composite airfoils include a polymer matrix composite (PMC).
14. The turbine engine of claim 1 wherein at least one of the first leading edge protector or the second leading edge protector is a metallic leading edge protector.
15. The turbine engine of claim 1 wherein the composite airfoils of the first stage of composite airfoils are fan blades and the composite airfoils of the second stage of composite airfoils are outlet guide vanes.
16. The turbine engine of claim 1 wherein the SPF is determined at a location between 20% and 80% of the first span length and the second span length, inclusive of endpoints.
17. The turbine engine of claim 1 wherein the first leading edge protector and the second leading edge protector each comprise a sheath.
18. The turbine engine of claim 1 wherein the first stage of composite airfoils has a first number of airfoils and the second stage of composite airfoils has a second number of airfoils and the first number is different than the second number.
19. The turbine engine of claim 1 wherein the first stage of composite airfoils and the second stage of composite airfoils are configured to rotate.
20. The turbine engine of claim 1 wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$).
21. The turbine engine of claim 1 wherein the FLL ranges from 6 cm to 19 cm and the SLL ranges from 1.5 cm to 4 cm.
22. The turbine engine of claim 1 wherein the FCL ranges from 24 cm to 77 cm and the SCL ranges from 9.3 cm to 31 cm.
23. The turbine engine of claim 1 wherein: the turbine engine is an indirect turbine engine comprising a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, and a speed reduction device driven by the turbine section for rotating the fan; and wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length, and the second airfoil extends spanwise between a second root and a second tip to define a second span length less than the first span length.
24. The turbine engine of claim 1 wherein the turbine engine is an indirect turbine engine comprising a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, and a speed reduction device driven by the turbine section for rotating the fan.
25. The turbine engine of claim 1 wherein the turbine engine is an indirect turbine engine comprising a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, and a speed reduction device driven

by the turbine section for rotating the fan.

26. A turbine engine comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a set of composite airfoils comprising a set of fan blades and a set of outlet guide vanes downstream from the set of fan blades, the set of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the LL is related to the CL by an airfoil protection factor (APF) $= \frac{(LL)}{(CL)}$, a first APF (APF1) for the set of fan blades is greater than or equal to 0.2 and less than or equal to 0.30 (0.2≤APF1≤0.3), and a second APF (APF2) for the set of outlet guide vanes is greater than or equal to 0.08 and less than or equal to 0.17 (0.08≤APF2≤0.17); wherein at least one of a fan blade of the set of fan blades or an outlet guide vane of the set of outlet guide vanes is secured to the rotor or the stator by a platform, the platform providing a flowpath boundary, the platform comprising: a structural body portion comprising a hollow body having a first side wall, a second side wall, a radially inner surface extending between said first and second side walls, and open ends, the structural body portion having a contour that matches that of an adjacent structural body portion; and a flowpath surface portion joined to the structural body portion, the flowpath surface portion defining a pair of frangible wings that extend laterally beyond the structural body portion.

27. The turbine engine of claim 26 wherein the APF1 relates to the APF2 by an expression: $\frac{APF1}{APF2}$ to define a stage protection factor (SPF), wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 (0.95≤SPF≤2.5).

28. The turbine engine of claim 26 wherein the composite airfoils in the set of composite airfoils extend spanwise between a root and a tip to define a span length, and the APF is determined at a location between 20% and 80% of the span length.

29. The turbine engine of claim 26 wherein the leading edge protector overlaps with the composite leading edge to define a sheath.

30. The turbine engine of claim 26 wherein the composite portion is formed from a polymer matrix composite (PMC).

31. The turbine engine of claim 26 wherein the leading edge protector is a metallic leading edge protector.

32. The turbine engine of claim 26 wherein the LL ranges from 6 cm to 19 cm for the set of fan blades and from 1.5 cm to 4 cm for the set of outlet guide vanes.

33. The turbine engine of claim 26 wherein the CL ranges from 24 cm to 77 cm for the set of fan blades and from 9.3 cm to 31 cm for the set of outlet guide vanes.

34. A turbine engine, comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a first stage of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge; a first leading edge protector receiving at least a portion of the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL); and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL); a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second

composite portion extending chordwise between a second composite leading edge and a second trailing edge; a second leading edge protector receiving at least a portion of the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL); and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL); wherein the FLL and the FCL are related to the SLL and the SCL by a stage protection factor (SPF), wherein the $SPF = \frac{(FLL / FCL)}{(SLL / SCL)}$, and the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$); wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length, and the second airfoil extends spanwise between a second root and a second tip to define a second span length less than the first span length; wherein the turbine engine further includes a platform for circumferential disposition between fan blades, the platform comprising: a platform wall with a radially outer surface which faces radially outwardly and a radially inner surface which faces radially inwardly, the platform wall being sloped with respect to a centerline with an increasing radius of the outer surface in an axially aft direction, forward, mid, and aft mounting lugs depend radially inwardly from the platform wall, and a wedge-shaped platform bumper depends radially inwardly from the inner surface of the platform wall axially forward of the mid mounting lug.
35. The turbine engine of claim 34 further comprising circumferentially curved forward stiffening ribs extending axially from the mid mounting lug to a forward edge of the platform bumper about where the platform bumper begins to depend radially inwardly from the inner surface of the platform wall.
36. The turbine engine of claim 35 wherein the circumferentially curved forward stiffening ribs are tapered down to the inner surface of the platform such that at any axial location, a first height of the circumferentially curved forward stiffening ribs is less than a second height of the platform bumper along an axially extending bumper length.
37. The turbine engine of claim 34 further comprising circumferentially curved aft stiffening ribs extending between the mid and aft mounting lugs.
38. The turbine engine of claim 34 wherein the platform wall further comprises a rectangularly shaped forward portion and a circumferentially curved aft portion.
39. The turbine engine of claim 38 wherein the circumferentially curved aft portion has pressure and suction side edges that have pressure and suction side airfoil shapes respectively.
40. The turbine engine of claim 34 wherein the first stage of composite airfoils and the second stage of composite airfoils include a polymer matrix composite (PMC).
41. The turbine engine of claim 34 wherein at least one of the first leading edge protector or the second leading edge protector is a metallic leading edge protector.
42. The turbine engine of claim 34 wherein the composite airfoils of the first stage of composite airfoils are fan blades and the composite airfoils of the second stage of composite airfoils are outlet guide vanes.
43. The turbine engine of claim 34 wherein the SPF is determined at a location between 20% and 80% of the first span length and the second span length, inclusive of endpoints.
44. The turbine engine of claim 34 wherein the first leading edge protector and the second leading edge protector each comprise a sheath.
45. The turbine engine of claim 34 wherein the first stage of composite airfoils has a first number of airfoils and the second stage of composite airfoils has a second number of airfoils and the first number is different than the second number.
46. The turbine engine of claim 34 wherein the first stage of composite airfoils and the second stage of composite airfoils are configured to rotate.
47. The turbine engine of claim 34 wherein the SPF is greater than or equal to 0.95 and less than or

equal to 2.5 (0.95≤SPF≤2.5).
48. A turbine engine comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a set of composite airfoils comprising a set of fan blades and a set of outlet guide vanes downstream from the set of fan blades, the set of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the LL is related to the CL by an airfoil protection factor (APF) = $\frac{(LL)}{(CL)}$, a first APF (APF1) for the set of fan blades is greater than or equal to 0.2 and less than or equal to 0.30 (0.2≤APF1≤0.3), and a second APF (APF2) for the set of outlet guide vanes is greater than or equal to 0.08 and less than or equal to 0.17 (0.08≤APF2≤0.17); wherein the turbine engine further includes a platform for circumferential disposition between fan blades of the set of fan blades, the platform comprising: a platform wall with a radially outer surface which faces radially outwardly and a radially inner surface which faces radially inwardly, the platform wall being sloped with respect to a centerline with an increasing radius of the outer surface in an axially aft direction, forward, mid, and aft mounting lugs depend radially inwardly from the platform wall, and a wedge-shaped platform bumper depends radially inwardly from the inner surface of the platform wall and is axially located forward of the mid mounting lug.
49. The turbine engine of claim 48 further comprising circumferentially curved forward stiffening ribs extending axially from the mid mounting lug to a forward edge of the platform bumper about where the platform bumper begins to depend radially inwardly from the inner surface of the platform wall.
50. The turbine engine of claim 49 wherein the circumferentially curved forward stiffening ribs are tapered down to the inner surface of the platform such that at any axial location, a first height of the circumferentially curved forward stiffening ribs is less than a second height of the platform bumper along an axially extending bumper length.
51. The turbine engine of claim 48 further comprising circumferentially curved aft stiffening ribs extending between the mid and aft mounting lugs.
52. The turbine engine of claim 48 wherein the platform wall further comprises a rectangularly shaped forward portion and a circumferentially curved aft portion.
53. The turbine engine of claim 52 wherein the circumferentially curved aft portion has pressure and suction side edges that have pressure and suction side airfoil shapes respectively.
54. The turbine engine of claim 48 wherein the APF1 relates to the APF2 by an expression: $\frac{APF1}{APF2}$ to define a stage protection factor (SPF), wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 (0.95≤SPF≤2.5).
55. The turbine engine of claim 48 wherein the composite airfoils in the set of composite airfoils extend spanwise between a root and a tip to define a span length and the APF is determined at a location between 20% and 80% of the span length.
56. The turbine engine of claim 48 wherein the leading edge protector overlaps with the composite leading edge to define a sheath.
57. The turbine engine of claim 48 wherein the composite portion is formed from a polymer matrix composite (PMC).
58. The turbine engine of claim 48 wherein the leading edge protector is a metallic leading edge protector.
59. The turbine engine of claim 48 wherein the LL ranges from 6 cm to 19 cm for the set of fan blades and from 1.5 cm to 4 cm for the set of outlet guide vanes.

60. The turbine engine of claim 48 wherein the CL ranges from 24 cm to 77 cm for the set of fan blades and from 9.3 cm to 31 cm for the set of outlet guide vanes.